

Efficient Estimation of Nonlinear DSGE Models

Eva F. Janssens[†] and Sean McCrory[‡]

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Abstract

This paper introduces a computationally efficient method for full-information estimation of nonlinear Dynamic Stochastic General Equilibrium (DSGE) models. In contrast to the traditional approach of treating model solution and filtering as separate steps, we integrate the solution directly into the filtering procedure. The method computes a sequence of local linear solutions at the best forecast of the current state of the economy, resulting in a time-varying linear state-space representation that enables likelihood evaluation via the time-varying Kalman filter. By avoiding both global solution methods and nonlinear filtering, the method reduces the computational burden of nonlinear estimation by several orders of magnitude while maintaining accuracy. We confirm this by benchmarking on the Diamond–Mortensen–Pissarides and textbook New Keynesian models. Applying the method to estimate a New Keynesian model with labor market frictions, we show that nonlinearities matter: monetary policy transmission is state-dependent, with the inflation–unemployment trade-off larger during expansions. We also show how the framework can be extended to incorporate stochastic volatility, regime switching, and heterogeneous agents.

Keywords: Bayesian inference, computational methods, Taylor projection, labor search

JEL classification codes: C63, C68, C32, E32, E52, J64

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Contact information:

[†] Eva F. Janssens: University of Michigan, Department of Economics. Email: evafj@umich.edu

[‡] Sean McCrory: Ohio State University, Department of Economics. E-mail: mccrory.65@osu.edu

1 Introduction

Nonlinearities play a fundamental role in macroeconomic dynamics, and recent empirical evidence increasingly underscores their importance. Several studies suggest that the Phillips curve is nonlinear, and a growing literature documents that the transmission of economic shocks is state-, size-, or sign-dependent.¹ While macroeconomic models often feature such nonlinearities, whether they meaningfully affect model predictions depends crucially on the choice of solution and estimation method. Global solution methods combined with nonlinear filters are seen as the most accurate way to estimate models with such nonlinearities, but their computational cost quickly becomes prohibitive: estimation can take days, weeks, or months.²

The methodological contribution of this paper is a generally applicable and computationally efficient approach for full-information estimation of nonlinear Dynamic Stochastic General Equilibrium (DSGE) models. Traditional methods treat model solution and likelihood evaluation (via filtering) as two different steps. In contrast, we integrate the model solution directly into the filtering step. The key insight is that, for estimation, the model solution need only be characterized in regions of the state space where the data suggest the economy is most likely to be. Instead of computing a globally accurate solution upfront, our method solves the model locally and updates the solution recursively during filtering.

Concretely, the method constructs a dynamic sequence of local linear solutions, embedding the resulting policy rules in a time-varying linear state-space system. Each local solution is fast to compute and accurate in a neighborhood of the current state. The overall approximation remains precise because each solution is used only for a single prediction step before being recomputed at the next forecast. The linear state-space form enables likelihood evaluation via the time-varying Kalman filter, which is both computationally efficient and optimal given the structure of the solution, and eliminates the need for simulation-based filters such as the particle filter.

Together, these features allow the method to sidestep the two main computational bottlenecks in nonlinear DSGE estimation – global model solutions and nonlinear filtering – without sacrificing accuracy. The approach applies to a broad class of DSGE models, including heterogeneous agent models, and captures key nonlinearities in a tractable way. Importantly, the dynamic implementation of local linear methods preserves state dependence and, despite the policy functions being linear, yields solutions that are not certainty equivalent.

¹For example, Benigno and Eggertsson (2024); Gitti (2024); Hazell, Herreno, Nakamura, and Steinsson (2022); Smith, Timmermann, and Wright (2023); Stock and Watson (2020) examine time variation in the Phillips curve; Tenreyro and Thwaites (2016) studies state dependence in monetary policy transmission; Barnichon, Debortoli, and Matthes (2022) analyzes sign dependence in government spending multipliers; and Ottonello and Winberry (2020) demonstrate state-dependence in firms’ distance to default.

²For example, in Gust, Herbst, López-Salido, and Smith (2017), a single solution and likelihood evaluation takes six minutes in serial. Since Bayesian samplers typically require 100,000 evaluations to characterize the posterior, this implies over a year of computation time.

We benchmark the proposed estimation method on two canonical models to assess its accuracy and computational efficiency. First, in the highly nonlinear Diamond-Mortensen-Pissarides (DMP) model,³ we use a global solution as the data-generating process and show that our method exceeds or matches the accuracy of the true global solution combined with either the Particle Filter (with finite particles) or the Extended Kalman Filter, but at a fraction of the computational cost. Second, we estimate the canonical three-equation New Keynesian (NK) model (Gali, 2015). Our method produces 100,000 posterior draws in under eight minutes—less than twice the time required to estimate the linearized model in Dynare,⁴ and several orders of magnitude faster than approaches based on particle filters and/or global solutions.⁵

Main Application We apply the proposed estimation framework to study state dependence in monetary policy transmission within an NK model with DMP labor market frictions. In search models, the flow surplus – the profitability of job creation – determines how shocks propagate through the economy (Ljungqvist and Sargent, 2017). We show that this mechanism extends to monetary policy shocks: in linearized models, the flow surplus is fixed, while in the nonlinear model it evolves endogenously, giving rise to state-dependent transmission. Our method captures this state dependence by solving the model locally along the filtered state path, allowing elasticities to vary across the state space. This makes the application well suited to illustrate how the method can estimate models with state-dependent transmission in a computationally efficient way. While estimation using global solution methods and nonlinear filters would require roughly three months to generate 100,000 posterior draws,⁶ our method estimates the model in approximately 2.5 hours.

The framework allows us to compute model-implied state-dependent impulse response functions, and we find substantial variation in the effects of monetary shocks on output, unemployment, and market tightness – by as much as a factor of two, driven by the underlying state of the economy. One policy-relevant implication is time variation in the Phillips Multiplier, which averages 0.15, consistent with Barnichon and Mesters (2021), but falls to 0.08 in recessions and rises toward 0.18 in expansions. We validate these model-based findings using reduced-form evidence from state-dependent local projections (Cloyne, Jordà, and Taylor, 2023). More broadly, our estimated NK-DMP model demonstrates that several nonlinear features highlighted in recent empirical work – such as state-dependent monetary transmission (Tenreyro and Thwaites, 2016) and a cyclical Phillips curve slope (Benigno and Eggertsson, 2024; Gitti, 2024) – can arise endogenously in standard macroeconomic models when solved and estimated nonlinearly, without relying on kinks or time-varying parameters. Our method makes such analysis computationally feasible.

³Diamond (1982), Mortensen (1982), and Pissarides (1985). See also Petrosky-Nadeau and Zhang (2017).

⁴Adjemian et al. (2024).

⁵An and Schorfheide (2007) report that particle filtering with second-order perturbation is about 200 times slower than Kalman filtering with first-order perturbation. Global solutions are at least 500 times slower than perturbation methods even for small models (Kollmann, Maliar, Malin, and Pichler, 2011).

⁶An optimistic estimate, based on Fernández-Villaverde and Rubio-Ramírez (2007) and Judd, Maliar, Maliar, and Valero (2014), suggests at least 10 seconds per draw for filtering and 60 seconds per model solution, implying approximately 11 weeks of runtime.

Extensions Beyond the baseline framework, we show how the method can be extended to accommodate two important classes of shock processes commonly used in DSGE models:⁷ (i) stochastic volatility and (ii) regime switching. For stochastic volatility, the method can be adapted by replacing the time-varying Kalman filter with an Extended Kalman Filter, or by discretizing the volatility process, adapting Farmer (2021). For regime-switching models, we include a Hamilton (1994) step directly in the filtering procedure. This extension nests, among others, models featuring rare disasters (Barro, 2009).

Finally, we extend the method to heterogeneous agent models. Unlike existing full-information approaches (Auclert, Bardóczy, Rognlie, and Straub, 2021; L. Liu and Plagborg-Møller, 2023; Winberry, 2018), which rely on perturbation techniques and impose certainty equivalence with respect to aggregate risk, our framework allows aggregate uncertainty to directly affect individual decision rules. This yields a richer characterization of household behavior, while remaining more computationally efficient than global solution methods such as Krusell and Smith (1998). Importantly, the approach preserves a time-varying linear state-space structure and avoids the need for nonlinear filters. The method also captures state dependence with respect to cross-sectional distributions, which has been shown to matter in, for example, heterogeneous firm models (Ottonello and Winberry, 2020). As a demonstration, we apply the method to a heterogeneous household model where aggregate conditions affect job loss and finding probabilities and a global solution is needed to capture the relevant precautionary-saving motives.

Related literature Our contribution to filtering and estimation is a novel framework for the full-information estimation of nonlinear DSGE models. In representative-agent settings, a key benchmark is Fernández-Villaverde and Rubio-Ramírez (2007), who introduce the bootstrap particle filter to economics. While widely applicable, the particle filter is computationally costly and suffers from the curse of dimensionality. Herbst and Schorfheide (2019) propose the tempered particle filter, which improves stability and reduces particle degeneracy, but is even more computationally intensive. Farmer (2021) introduces a discretization filter, which is linear hence fast, but the approach is limited to small state spaces due to its reliance on tensor grids. Winschel and Krätzig (2010) mitigate some of these issues using sparse grids and parallelization, though at the cost of high implementation complexity, and still the computational burden remains substantial. All of these methods apply a filter to a model solution obtained in a separate step. In contrast, our method integrates model solution directly into the filtering step, solving the model dynamically as the filter proceeds and avoiding the need for simulation-based filters. It is accurate, computationally efficient, and feasible on a standard desktop without relying on parallelization. Also related is Childers, Fernández-Villaverde, Perla, Rackauckas, and Wu (2022), who focus on Bayesian parameter sampling. Their approach avoids filtering altogether but applies only to first- or second-order perturbation solutions, which may yield imprecise approximations in the presence of nonlinear

⁷See, e.g., Fernández-Villaverde and Levintal (2018); Fernández-Villaverde and Rubio-Ramírez (2007); Foerster, Rubio-Ramírez, Waggoner, and Zha (2016); Z. Liu, Waggoner, and Zha (2011).

dynamics (Den Haan and De Wind, 2012; Judd, Maliar, and Maliar, 2017). Finally, in the context of heterogeneous agent models, existing computationally feasible full-information estimation procedures rely on perturbation methods (Auclert et al., 2021; L. Liu and Plagborg-Møller, 2023; Winberry, 2018), which impose certainty equivalence with respect to aggregate risk. Relative to these, our framework provides an efficient estimation method that accommodates nonlinearities in the aggregate dynamics.

In terms of solution method, we build on Den Haan, Kobielarz, and Rendahl (2015), Levintal (2018), and Mennuni, Rubio-Ramírez, and Stepanchuk (2024). Levintal (2018) propose Taylor Projection, which computes a (linear or nonlinear) local policy rule optimal at a specific point in the state space. Fernández-Villaverde and Levintal (2024) apply Taylor Projections to solve a heterogeneous agent model. We adopt the Taylor Projection approach for its speed, and apply it dynamically within our filtering framework, computing a new local policy rule at each forecasted state. None of these papers consider full-information estimation, which is our focus.

Our main application builds on the integration of New Keynesian and search-and-matching models (Diamond, 1982; Mortensen, 1982; Pissarides, 1985), extending work such as Christiano, Eichenbaum, and Trabandt (2016), Thomas (2011), and Galí, Smets, and Wouters (2012). The model is closest in structure to Krause, Lopez-Salido, and Lubik (2008), but differs in a key respect: whereas they use log-linearization, which fixes flow profits at their steady-state level, we solve the model nonlinearly, allowing flow profits to vary endogenously. This generates state-dependent monetary transmission, which is our main contribution relative to this literature. We view this mechanism as complementary to alternative explanations based on pricing frictions (Dotsey, King, and Wolman, 1999; Gertler and Leahy, 2008; Harding, Lindé, and Trabandt, 2022). Our application also relates to recent work on the slope of the Phillips curve (Benigno and Eggertsson, 2023; Gitti, 2024), providing evidence that the slope is steeper in tight labor markets. While Benigno and Eggertsson (2023) and Gitti (2024) impose explicit kinks to generate this feature, we show that it can arise endogenously from solving the model nonlinearly.

Outline The rest of this paper is structured as follows. Section 2 presents the method. Section 3 demonstrates the performance of this method using Monte-Carlo simulations from the Diamond-Mortensen-Pissarides model and the evaluation of the three-equation New-Keynesian model. Section 4 presents the main application. Section 5 presents the quantitative model results and supporting reduced-form evidence. Section 6 discusses extensions with stochastic volatility, regime switching and heterogeneous households. Section 7 concludes.

2 Local Linear Approximations and the Time-Varying Kalman Filter

This section presents our main methodological contribution: a full-information estimation framework for nonlinear DSGE models based on local linear approximations. The key innovation is to

integrate model solution directly into the filtering process, enabling fast and accurate likelihood evaluation without solving the model globally.

2.1 A General Dynamic Stochastic General Equilibrium (DSGE) model

A standard Dynamic Stochastic General Equilibrium (DSGE) model can be represented as follows (Schmitt-Grohé and Uribe, 2004):

$$\mathbb{E}_t [f(y_{t+1}, y_t, x_{t+1}, x_t)] = 0, \quad (1)$$

where y_t represents the control variables, and the state variables x_t can be partitioned into endogenous variables $x_{1,t}$ and exogenous variables $x_{2,t}$. We assume the exogenous variables follow a stationary VAR(1):

$$x_{2,t+1} = Ax_{2,t} + B\eta_{t+1}, \quad (2)$$

where η_t is a vector of Gaussian exogenous shocks. Endogenous variables $x_{1,t+1}$ depend on x_t through an unknown nonlinear mapping. Hence, a solution to (1) is characterized by two implicit functions,

$$x_{1,t+1} = h(x_t)$$

which implies the states x_t evolve as

$$x_{t+1} = \begin{pmatrix} h(x_t) \\ Ax_{2,t} \end{pmatrix} + \begin{pmatrix} 0 \\ B \end{pmatrix} \eta_{t+1},$$

and the policy function:

$$y_t = g(x_t).$$

A and B are assumed to be known functions of structural parameters (that may have to be estimated), while g and h are the unknown functions that we wish to solve for.

The functions g and h are typically solved using either perturbation methods, which are accurate near the steady state, or projection methods, which are globally accurate but computationally intensive. Standard estimation approaches separate model solution from likelihood evaluation. In contrast, our method integrates the solution into the filter, constructing locally accurate solution methods around the best current forecast of the state of the economy. Our preferred implementation builds on the local approximation method called Taylor projections (Fernández-Villaverde and Levintal, 2018; Levintal, 2018), which we find to be significantly more computationally efficient than alternative methods. However, the filtering framework we describe is more general and can also be applied to other local linearization techniques, such as Dynamic Perturbation (Mennuni et al., 2024).

2.2 Taylor Projection (Levintal, 2018)

We now outline the first-order Taylor projection method, following Levintal (2018). The method consists of four steps. In the first step, the policy function $g(x_t)$ and law of motion $h(x_t)$ are approximated around a generic point x using linear functions:⁸

$$\tilde{g}(x_t; \Theta) = \Theta_1 + \Theta_2 x_t \quad (3)$$

$$\tilde{h}(x_t; \Theta) = \Theta_3 + \Theta_4 x_t \quad (4)$$

Note this implies the following approximate controls $\tilde{y}_t, \tilde{y}_{t+1}$, and future states $\tilde{x}_{t+1}$

$$\tilde{y}_t = \Theta_1 + \Theta_2 x_t \quad (5)$$

$$\tilde{x}_{t+1} = \begin{pmatrix} \Theta_3 \\ 0 \end{pmatrix} + \begin{pmatrix} \Theta_4 \\ A \end{pmatrix} x_t + \begin{pmatrix} 0 \\ B \end{pmatrix} \eta_{t+1} \quad (6)$$

$$\tilde{y}_{t+1} = \Theta_1 + \Theta_2 \tilde{x}_{t+1} \quad (7)$$

In step two, the residual function $R(x_t, \Theta)$ is constructed by substituting $\tilde{y}_t, \tilde{y}_{t+1}, \tilde{x}_{t+1}$ as defined above into the equilibrium condition (1):

$$R(x_t, \Theta) = \mathbb{E}_t [f(\tilde{y}_{t+1}, \tilde{y}_t, \tilde{x}_{t+1}, x_t)] \approx 0. \quad (8)$$

The objective is to find Θ such that the residual function is equal to zero around x . To do so, in step three, the residual function is approximated using a Taylor series expansion around a specific point x , retaining terms up to first order:

$$R(x_t, \Theta) \approx R(x, \Theta) + \left. \frac{\partial R(x_t, \Theta)}{\partial x_t} \right|_{x_t=x} (x_t - x). \quad (9)$$

Finally, in step four, one solves for the optimal Θ^* such that the Taylor series satisfies:

$$\begin{aligned} R(x, \Theta^*) &= 0 \\ \left. \frac{\partial}{\partial x_t} R(x_t, \Theta^*) \right|_{x_t=x} &= 0. \end{aligned}$$

Note that x may be a vector, so our compact notation for the first-order derivatives intends to capture all partial derivatives. Solving this system at a particular state x gives a set of state-dependent parameters Θ_x , yielding an approximate solution $\tilde{g}(x_t, \Theta_x)$ and $\tilde{h}(x_t, \Theta_x)$ around x . In our following examples, we will make more explicit how the Taylor projection steps work in practice.

⁸Note, we could have written $\tilde{g}(x_t; \tilde{\Theta}) = \tilde{\Theta}_1 + \tilde{\Theta}_2(x_t - x)$ and $\tilde{h}(x_t; \tilde{\Theta}) = \tilde{\Theta}_3 + \tilde{\Theta}_4(x_t - x)$ as in Levintal (2018). This is an equivalent representation with $\Theta_1 = \tilde{\Theta}_1 - \tilde{\Theta}_2 x$ and so on, but we chose this to simplify the filtering notation in the next sections.

No Certainty Equivalence. Unlike standard first-order perturbation methods, the Taylor projection does not impose certainty equivalence. Because the residual function $R(x_t, \Theta)$ contains expectations over future shocks, the resulting decision rules internalize risk, despite being linear.

Dynamic Taylor Projection. To improve accuracy away from a single approximation point, we recompute the Taylor projection dynamically along the state path – similar to the Dynamic Perturbation method of Mennuni et al. (2024). This yields a sequence of solutions $\tilde{g}(x_t, \Theta_x)$ and $\tilde{h}(x_t, \Theta_x)$, accurate locally around x . As long as the approximate solutions computed at $x = x_t$ are accurate at x_{t+1} , the mistakes made by the dynamic implementation of the solution method will be small.

Avoiding Numerical Integration. The residual function involves expectations, typically approximated using numerical integration. However, when policies are linear in logs and shocks are Gaussian, expectations can oftentimes be computed analytically, including in several of our applications.⁹ We give additional details on this approach for our applications in the Appendix. This yields an exact nonlinear system, improving both speed and accuracy. It eliminates integration error, accelerates residual evaluation, and produces exact Jacobians for nonlinear solvers. While not necessary, this structure is common in models designed for log-linearization, making it a practical advantage in many applications.

2.3 Time-Varying Kalman Filter

Next, we discuss how the local linear solution method can be incorporated into a filtering framework. Consider a standard linear state space model:

$$y_t^{\text{obs}} = Zx_t + d + \varepsilon_t \quad (10)$$

$$x_{t+1} = Tx_t + c + R\eta_{t+1} \quad (11)$$

with ε_t i.i.d. measurement error with variance-covariance matrix H , and η_t i.i.d. with variance-covariance Q . y_t^{obs} is the set of observed variables, and x_t are the states, which are assumed unobserved, and correspond one-to-one to the states in the DSGE model in Equation (1). Also innovation vector η_t corresponds exactly to η_t in our general notation above.

It is well-known that the first-order perturbation solution of DSGE models can be mapped into a linear state space form, where the system matrices Z , d , H , Q , T , c and R depend on structural model parameters. This means that its parameters can be estimated using standard Kalman filtering approaches. For a textbook treatment, consider Herbst and Schorfheide (2016). For a standard model that is log-linearized around the steady state, the system matrices are constant and do not vary over time.

⁹For a Normal random variable $\mathbb{E}[e^{cx}] = e^{c\mu + \frac{1}{2}c^2\sigma^2}$.

Key to our method is the insight that when a model is solved using a local linear approximation method such as Levintal (2018)’s Taylor Projection method outlined above, the local linear model solution can be mapped into a *time-varying* state space model, and the model’s states can be filtered using a time-varying Kalman Filter.

Assume the Taylor Projection method around a specific point x is used to obtain an approximation $y_t = \Theta_{1,x} + \Theta_{2,x}x_t$ and $x_{1,t+1} = \Theta_{3,x} + \Theta_{4,x}x_t$, where the subscript x indicates that these are the solution matrices based on the Taylor projection around a specific point x . Furthermore, assume that the observed variables y_t^{obs} are linear functions of x_t or y_t , or can be well approximated as such. Given that our model solution is characterized by a set of linear equations, we can map the model solution into a linear state space form:

$$y_t^{\text{obs}} = Z_x x_t + d_x + \varepsilon_t \quad (12)$$

$$x_{t+1} = T_x x_t + c_x + R\eta_{t+1} \quad (13)$$

where T_x , c_x and R follow directly from Equation (6). The observation equation matrices Z_x , d_x may contain elements of the matrices Θ_x of the local approximation of the model solution as well as other model parameters depending on the observables of interest. For the models we consider in the main part of this paper, the variance-covariance matrices H and Q , as well as shock selection matrix R do not have state-dependence when using this solution method, so these matrices do not need to be indexed by x . In the extensions in Section 6, we discuss how to allow for nonlinearities of this form.

The Kalman filter is valid and optimal, even when the system matrices Z , d , T and c are time-varying, as long as this time-dependence is fully known when conditioning on the information set $\mathcal{Y}_{t-1} = \{y_{t-1}^{\text{obs}}, y_{t-2}^{\text{obs}}, \dots\}$. This follows from the recursive definition of the Kalman filter.¹⁰ This allows us to replace the subscript x in Equations (12)-(13) by $\hat{x}_{t|t-1}$, our prediction for the state x_t using all information up to and including time $t - 1$, and implies the filtering procedure outlined in Algorithm 1. Algorithm 2 in Appendix A outlines the Smoothing algorithm.

In words, what Algorithm 1 does is it computes a local linear approximation of the policy functions g and law of motion h at the best forecast for the current state x_t , denoted $\hat{x}_{t|t-1}$ based on all data up to and including $t - 1$. We can map this linear approximate model solution into the linear state space form of Equations (12)-(13), and use the Kalman filter to obtain a prediction for the next state, $\hat{x}_{t+1|t}$. We then repeat the procedure, by computing a local linear model solution around $\hat{x}_{t+1|t}$, mapping this model solution into state space form, and obtaining a new prediction $\hat{x}_{t+2|t+1}$, etc. The Kalman Smoother in Algorithm 2 uses the same system matrices as Algorithm 1, so, attractively, does not require re-solving the model.

¹⁰This is a well-established result; see, for example, Durbin and Koopman (2012) and Hamilton (1994).

Algorithm 1 Time-Varying Kalman Filtering with Taylor Projections

Initialize: Set $t = 0$, start x_0 in the steady state, and initialize covariance $P_{0|0}$.

while $t \leq T$ **do**

Taylor Projection Step:

 Solve the model using Taylor Projections at $\hat{x}_{t|t-1}$, obtaining the time-varying system:

$$y_t^{\text{obs}} = Z_{\hat{x}_{t|t-1}} x_t + d_{\hat{x}_{t|t-1}} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H) \quad (14)$$

$$x_{t+1} = T_{\hat{x}_{t|t-1}} x_t + c_{\hat{x}_{t|t-1}} + R\eta_{t+1}, \quad \eta_t \sim \mathcal{N}(0, Q) \quad (15)$$

Update Step:

 Compute the Kalman gain:

$$F_t = Z_{\hat{x}_{t|t-1}} P_{t|t-1} Z'_{\hat{x}_{t|t-1}} + H \quad (16)$$

$$K_t = T_{\hat{x}_{t|t-1}} P_{t|t-1} Z'_{\hat{x}_{t|t-1}} F_t^{-1} \quad (17)$$

 Update the state estimate using the new observation y_t :

$$v_t = y_t^{\text{obs}} - Z_{\hat{x}_{t|t-1}} \hat{x}_{t|t-1} - d_{\hat{x}_{t|t-1}} \quad (18)$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + P_{t|t-1} Z'_{\hat{x}_{t|t-1}} F_t^{-1} v_t \quad (19)$$

 Update the covariance:

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} Z'_{\hat{x}_{t|t-1}} F_t^{-1} Z_{\hat{x}_{t|t-1}} P_{t|t-1} \quad (20)$$

Prediction step:

 Compute the predicted state estimate:

$$\hat{x}_{t+1|t} = T_{\hat{x}_{t|t-1}} \hat{x}_{t|t} + c_{\hat{x}_{t|t-1}} \quad (21)$$

 Compute the predicted covariance:

$$P_{t+1|t} = T_{\hat{x}_{t|t-1}} P_{t|t} T'_{\hat{x}_{t|t-1}} + RQR' \quad (22)$$

Advance time: Set $t \leftarrow t + 1$ and repeat.

end while

2.4 Comparison to Existing Nonlinear Full-Information Estimation Methods

Full-information estimation of nonlinear DSGE models typically proceeds in two steps: first, solving the model to characterize policy functions $\hat{g}$ and laws of motion $\hat{h}$; second, evaluating the likelihood via filtering. Our method departs from this separation by integrating the solution directly into the filtering process. Below, we compare our approach to existing methods along both dimensions.

Alternative solution methods The literature distinguishes between two main classes of solution techniques. The first is perturbation around the non-stochastic steady state. First-order (linear) perturbations are computationally efficient and map directly into a linear state-space form, enabling

straightforward estimation via the Kalman filter. However, linear perturbation imposes certainty equivalence, which ignores the role of risk, assume constant elasticities to aggregate shocks, and may yield inaccurate approximations away from the steady state (Judd et al., 2017). Higher-order perturbations (Judd, 1998) address some of these issues but are more complex, require pruning for stability (Andreasen, Fernández-Villaverde, and Rubio-Ramírez, 2018), and still rely on a single expansion point.

The second class comprises global solution methods, such as those based on function approximation or projection techniques (Judd et al., 2014). These methods are accurate across the entire state space but are substantially more computationally intensive. For example, Gust et al. (2017) report several minutes for a single model solution, and Kollmann et al. (2011) find global methods to be approximately 500 times slower than perturbation techniques. Our method avoids computing a global solution by constructing fast, locally accurate approximations via Taylor projection.

Alternative filtering methods Once a nonlinear solution is obtained, likelihood evaluation requires a nonlinear filter. A common approach is the bootstrap particle filter (Fernández-Villaverde and Rubio-Ramírez, 2007), which is accurate with sufficient particles but suffers from the curse of dimensionality: the number of particles required grows exponentially with the dimension of the state space (Bengtsson, Bickel, and Li, 2008). This makes even small models computationally expensive. The tempered particle filter (Herbst and Schorfheide, 2019) improves stability but increases the computational burden further by requiring more function evaluations. Alternatives such as the discretization filter (Farmer, 2021) are limited to low-dimensional settings due to reliance on tensor grids.

Another alternative is the extended Kalman filter (EKF), which linearizes the system around a forecasted path. While similar in spirit to our approach, EKF differs in that it relies on linearization after solving a nonlinear model. This distinction is critical: our method solves the model locally as part of the filtering process. We provide a more extensive comparison with EKF below.

The most accurate existing approach combines a global solution with a high-quality particle filter (e.g., Winschel and Krätzig, 2010), but this quickly becomes computationally prohibitive for medium-scale or larger models. For instance, Gust et al. (2017) estimate a medium-scale New Keynesian model using this approach, with computation times adding up to a full year. In contrast, our method provides a computationally efficient alternative that avoids both global solutions and nonlinear filters, without compromising accuracy.

Comparison to EKF The EKF operates by linearizing a nonlinear state-space model around the current estimate of the state. Consider a general nonlinear state-space system:

$$y_t^{\text{obs}} = Z(x_t) + \varepsilon_t, \quad (23)$$

$$x_{t+1} = T(x_t) + R\eta_{t+1}, \quad (24)$$

where ε_t and η_t are zero-mean innovations with covariances H and Q , respectively. Suppose the model is solved globally so that the equilibrium is represented by known policy functions $y_t = g(x_t)$ and law of motion $x_{1,t+1} = h(x_t)$. These functions define the mappings $Z(x_t)$ and $T(x_t)$ in the state-space model.

The EKF applies a first-order Taylor expansion of $Z(\cdot)$ and $T(\cdot)$ around the current state estimates:

$$y_t^{\text{obs}} = \dot{Z}_t x_t + d_t + \varepsilon_t, \quad d_t = Z(\hat{x}_{t|t-1}) - \dot{Z}_t \hat{x}_{t|t-1}, \quad (25)$$

$$x_{t+1} = \dot{T}_t x_t + c_t + R\eta_{t+1}, \quad c_t = T(\hat{x}_{t|t}) - \dot{T}_t \hat{x}_{t|t}, \quad (26)$$

where $\dot{Z}_t$ and $\dot{T}_t$ denote Jacobians evaluated at the current state estimates.

The key conceptual difference between the EKF and the TP+TV-KF approach lies in where the linearization occurs within the estimation process. In the EKF, linearizing $Z(x_t)$ and $T(x_t)$ amounts to linearizing the global policy functions $g(x_t)$ and $h(x_t)$ around some state forecast x , using a Taylor expansion of the global policy rule g and law of motion h :

$$\hat{h}(x_t) = h(x) + \dot{h}(x)(x_t - x),$$

$$\hat{g}(x_t) = g(x) + \dot{g}(x)(x_t - x).$$

The resulting linearized policies $\hat{h}$ and $\hat{g}$ generally do not satisfy the equilibrium conditions in (1), because the implied value

$$\hat{y}_{t+1} = \hat{g}(\hat{h}(x_t), Ax_{2,t} + B\eta_{t+1})$$

typically differs from the true value $g(h(x_t), Ax_{2,t} + B\eta_{t+1})$. In other words, a first-order Taylor expansion of an optimal policy rule around a point x is, in general, no longer optimal and violates the equilibrium conditions. In contrast, the TP method imposes that the policy is linear and solves the equilibrium conditions in a neighborhood around the current state forecast. As a result, TP+TV-KF enforces model-consistent dynamics by construction. Hence, the EKF and TP+TV-KF typically yield different linearized systems and therefore filtering performance.

3 Two Canonical Examples

In the first part of this section, we compare the performance of our proposed estimation framework to existing methods using simulations of the Diamond-Mortensen-Pissarides model. This model

is known to have significant nonlinear dynamics (Petrosky-Nadeau and Zhang, 2017), making it a natural laboratory for the proposed method. In the second part of this section, we illustrate how to use the method to estimate the textbook three-equation New Keynesian model, another model known to be poorly approximated using standard perturbation approaches (Judd et al., 2017). We apply the estimation method to US data, and focus on differences between the linearized and nonlinear model estimates, as well as computation time.

3.1 Diamond-Mortensen-Pissarides model

We briefly describe the model environment, as the model is standard. The only non-standard assumption is the one made with respect to the timing of separations and hires. Following the literature on labor search in NK models (Christiano et al., 2016), we assume workers who lose their jobs at the beginning of the period can search within the period for a new job. This assumption is typically justified as a way of making low-frequency (quarterly) NK models conform with labor search where most unemployment spell last only a few weeks.¹¹

Firms maximize the present discounted value of profit taking wages as given

$$\mathbb{E}_0 \left[\sum_{t=1}^{\infty} \beta^t \left\{ (Z_t - W_t) N_t - \kappa V_t \right\} \right]$$

subject to

$$N_t = (1 - \delta)N_{t-1} + q_t V_t,$$

where the flow payoff is output $Z_t N_t$, net of wage expenditures $W_t N_t$ and vacancy posting costs $\kappa V_t(j)$, δ is the job destruction rate, and q_t the vacancy filling rate. In equilibrium, the Nash wage is given by $W_t = \eta Z_t + (1 - \eta)v$, where η is the worker's bargaining weight, and v is the outside option of the worker. This leads to job creation condition

$$\frac{\kappa}{q_t} = (1 - \eta)(Z_t - v) + (1 - \delta)\beta E_t \left[\frac{\kappa}{q_{t+1}} \right], \quad (27)$$

which is the sole equilibrium condition in the model. We assume productivity is stochastic, and log productivity, denoted z_t , follows an AR(1):

$$z_{t+1} = \rho_z z_t + \varepsilon_{t+1}.$$

Model Solution The DMP model above has one state (TFP, z_t) and one policy to solve for: market tightness $\theta_t = g(z_t)$. Hence, mapping this back to our previous notation, our state $x_t = z_t$ and our

¹¹According to the estimate of Ahn and Hamilton (2022), two-thirds of workers exit unemployment within the first quarter of an unemployment spell.

control $y_t = \theta_t$. In the local linear approximation, we approximate the policy function by:

$$\theta_t = \Gamma_{1,z} + \Gamma_{2,z}z_t$$

which implies $\theta_{t+1} = \Gamma_{1,z} + \Gamma_{2,z}\rho_z z_t + \Gamma_{2,z}\varepsilon_{t+1}$ by iterating the law of motion for z_t one period forward.

The residual functions for the DMP model, used to solve for the local policy parameters, follow directly from the job creation condition in Equations (27), and are given by:

$$R(z, \Gamma_z)|_{z_t=z} = \frac{\kappa}{q(\theta_t)} - (1-\eta)(e^z - b) - \beta(1-\delta)\mathbb{E}_t \left[\frac{\kappa}{q(\theta_{t+1})} \right]$$

$$\frac{\partial R(z_t, \Gamma_z)}{\partial z_t}|_{z_t=z} = \frac{-\kappa q'(\theta_t)}{q(\theta_t)^2} \Gamma_{2,z} - (1-\eta)e^z + \beta(1-\delta)\rho_z \Gamma_{2,z} \rho_z \mathbb{E}_t \left[\frac{\kappa q'(\theta_t)}{q(\theta_t)^2} \right]$$

This implies two equations in two unknowns $\Gamma_{1,z}, \Gamma_{2,z}$, which can be solved for numerically.

Figure 1: Comparison of log-linearization, global model solution, and dynamic Taylor projection for the DMP model.

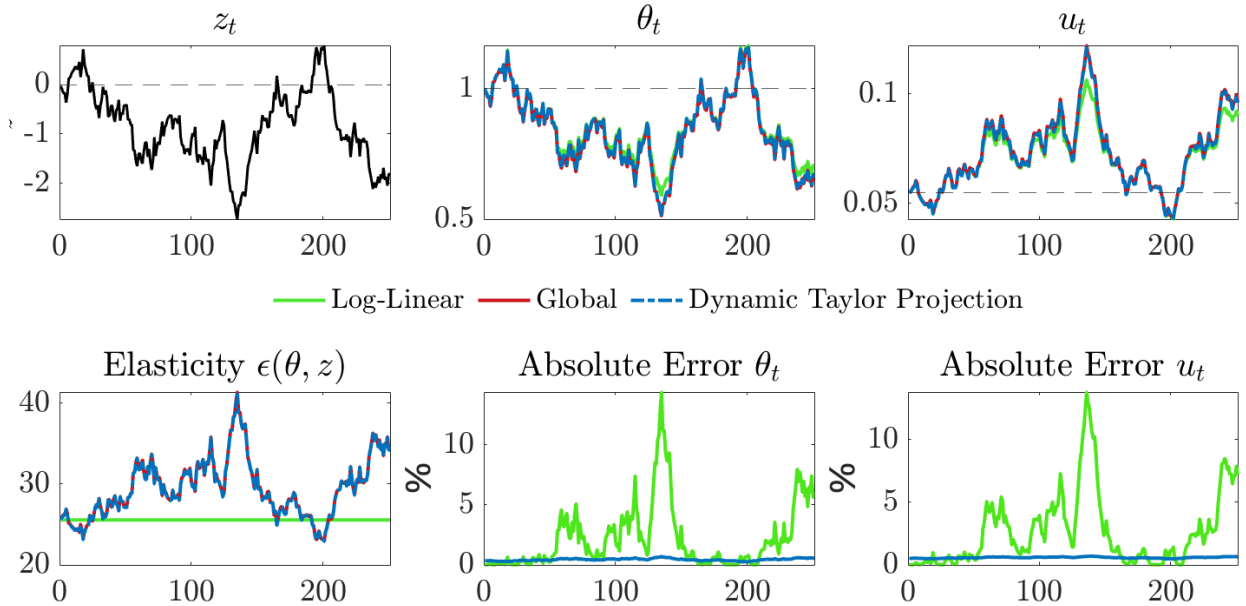


Figure 1 shows how a global model solution, approximating the unknown policy function by projection onto a set of polynomial basis functions, compares to a dynamic Taylor projection that computes the Taylor Projection solution at each point of the simulation path, and how these both compare to a log-linear model solution. The top row shows the simulated path for TFP, which is the same for all three methods, as well as the implied market tightness and unemployment rate. As can be seen, the simulation from the global model solution and dynamic Taylor projection follow each other closely. The log-linear solution deviates considerably, as measured by the two panels

in the bottom row, showing the absolute error compared to the global model solution in market tightness and the unemployment rate, which can be above 10% for the log-linear solution, while for the dynamic Taylor projection, is always low and close to the global solution.

The first panel in the second row shows that the local linear model solution also accurately tracks a key statistic of the DMP model: time-variability in the elasticity of market tightness and productivity. In a log-linear model solution, this elasticity is constant, since risk does not affect elasticities. Note that even in a second-order perturbation, this elasticity would be constant (Schmitt-Grohé and Uribe, 2004), because uncertainty only shifts the constant in the decision rule in a second-order perturbation, and does not affect elasticities.

State-Space Representation To make explicit how the Taylor projection solution maps into a linear state space model, assume we observe market tightness and wages, with measurement errors. Our local linear model solution around z is given by:

$$\theta_t = \Gamma_{1,z} + \Gamma_{2,z}z_t$$

and $w_t = \log W_t$ follows as:

$$w(z_t) \approx w(z) + \frac{\partial w(z)}{\partial z}(z_t - z)$$

This implies that the state-space representation follows as:

$$\begin{pmatrix} \theta_t^{\text{obs}} \\ w_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} \Gamma_{1,z} \\ \ln(\eta e^z + (1-\eta)b) - \frac{z\eta e^z}{\eta e^z + (1-\eta)b} \end{pmatrix} + \begin{bmatrix} \Gamma_{2,z} \\ \frac{\eta e^z}{\eta e^z + (1-\eta)b} \end{bmatrix} z_t + \underbrace{\varepsilon_t}_{\text{meas. error} \sim N(0,Q)}$$

$$z_{t+1} = \rho_z z_t + \eta_{t+1}, \quad \eta_t \sim (0, \sigma_z^2).$$

We can then use Algorithm 1 to filter the underlying states and evaluate the likelihood of this model.

Solution Accuracy: Euler Errors To evaluate the accuracy of the solution method, we compute Euler errors, which measure the extent to which the policy rule approximates the decision implied by the model's equilibrium condition, in this case the job creation condition (27). The Taylor projection solution satisfies the equilibrium condition exactly at the forecast point $z_{t|t-1}$. Thus, the relevant test is to apply the same policy rule to the subsequent forecasted state, $z_{t+1|t}$. Specifically, at each period t , we use the policy rule computed at $z_{t|t-1}$ to evaluate the expected Euler error implied by the equilibrium condition at future states $z_{t+1} \sim N(z_{t+1|t}, P_{t+1|t})$. Based on a simulation of $T = 10,000$ periods from a globally solved model with parameters reported in Table 1, we find that the average (over time) expected Euler error is -4.51 , with 99% of errors below -3.69 . This

indicates that the policy rule remains highly accurate even when applied one step ahead, before recomputing the solution at the next forecast.

Monte Carlo In this subsection, we present a simulation experiment based on 1,000 simulated datasets for two different lengths: $T = 200$, which is comparable to the length of a typical dataset used in macroeconomics, and $T = 1000$ to provide some insight into asymptotic behavior. The experiment uses the DMP model presented above.

The experiment first solves the DMP model using a global solution method and generates simulated data from this solution. We assume we observe market tightness with measurement error which has a standard error equal to 25% of the standard deviation of the true underlying variable. Then, we apply four different solution plus filtering approaches:

1. Linear perturbation with a standard Kalman filter (KF).
2. Global solution method with a particle filter (PF).
3. Taylor Projection (TP) with the time-varying Kalman filter (TP+TV-KF) in Algorithm 1.
4. Global solution method with Extended Kalman filter (EKF)

We fix the structural model parameters to their true values (see Table 1 notes), and our goal is to assess how well each method recovers the unobserved states from the simulated data. Table 1 evaluates performance by comparing the correlation of the filtered states with the true states, along with the root mean squared error (RMSE). Additionally, the table reports computation times for each approach.

Focusing on computation time, for $T = 1000$, Table 1 shows that TP+TV-KF is only about three times slower than Linear+KF, while Global+PF is several orders of magnitude slower than TP+TV-KF. Note that the DMP model, given that it has only one state, is quite easy to solve, even with global methods. In terms of accuracy, for low T , TP+TV-KF consistently outperforms both the global model + PF and the linear model + KF, across all metrics. For larger T , Global+PF with a sufficient number of particles does improve and approaches the accuracy of TP+TV-KF, but remains dominated. EKF and TP+TV-KF are comparable. We conclude that the combination of Taylor Projections and the Time-Varying Kalman Filter is an accurate and computationally efficient solution and filtering approach for this model.

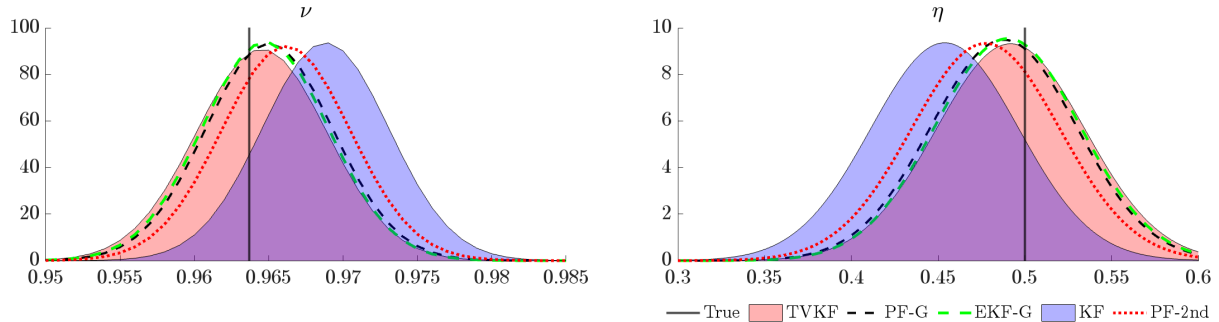
Comparing Posteriors Next, this subsection compares posteriors when the above-described filters are used in a Bayesian estimation routine with a Random Walk Metropolis Hastings (RWMH) algorithm for the posterior draws. We simulate a data set of length $T = 500$ from the global model solution parameterized as in the previous subsection. We assume we observe both market

Table 1: Comparing Solution and Filtering Methods for the DMP model. MSE is scaled by the (unconditional) variance of the variable.

	Linear + KF	Global + EKF	Global + PF			TP + TV-KF
Particles:			200	2,000	20,000	
T = 200						
Correlation Market Tightness	0.959	0.962	0.950	0.951	0.952	0.962
Correlation TFP	0.954	0.961	0.949	0.951	0.951	0.961
RMSE Market Tightness	0.199	0.192	0.216	0.213	0.213	0.192
RMSE TFP	0.226	0.138	0.156	0.154	0.153	0.138
Computation time (sec)	0.001	0.101	0.148	0.564	4.523	0.007
T = 1,000						
Correlation Market Tightness	0.964	0.968	0.965	0.966	0.967	0.968
Correlation TFP	0.949	0.967	0.965	0.966	0.966	0.967
RMSE Market Tightness	0.186	0.175	0.181	0.178	0.178	0.175
RMSE TFP	0.261	0.167	0.173	0.170	0.169	0.167
Computation time (sec)	0.008	0.109	0.359	2.487	22.62	0.032

Note: Calibration: $r = 0.04$, $\beta = \left(\frac{1}{1+r}\right)^{1/4}$, $\alpha = 0.7$, $\rho_z = 0.985$, $\sigma_z = 0.0015$, $\eta = 0.5$, $b = 0.94$, $n_{ss} = 0.945$, $q_{ss} = 0.7$, $\theta_{ss} = 1$.

Figure 2: Comparing posteriors for ν (outside option) and η (bargaining power) for the time-varying Kalman filter of Algorithm 1 (pink area), the Particle Filter + global model solution (black dash), Extended Kalman filter with global model solution (green dash), the linearized model with Kalman filter (blue area), and a second order perturbation with particle filter (red dotted).



tightness and wages, both with measurement error. For market tightness, the standard deviation of the measurement error equals 30% of the standard deviation of the simulated data series, and for wages 50%.¹² In addition to the four estimation routines above, we also add a second-order perturbation of the DMP model, estimated using the particle filter.¹³ For each method, we obtain 100,000 draws from the RWMH sampler.

¹²We need enough measurement error for the particle filter to be stable.

¹³We use the standard implementation in Dynare.

Focusing on economic parameters, Figure 2 shows the posterior distribution for the outside option ν and the bargaining power η . Both have a uniform prior $U[0, 1]$. For both parameters, the Kalman filter estimates are biased and the posterior distribution is shifted. The posterior distributions obtained from TP+TV-KF, Global+PF and Global+EKF closely align, and are close to the true value. A second-order perturbation filtered with particle filtering (red dotted line) only brings us half way from the biased linearized model parameter estimate to the true parameter, and hence is still biased, albeit less than the linear model.

3.2 Revisiting the Three-Equation New Keynesian Model

Our next example is the canonical three-equation New Keynesian model in Gali (2015). In brief, households are infinitely lived and maximize expected discounted utility, facing stochastic discount factor shocks d_t that follow an AR(1) process. They consume, save, and supply labor. Final goods are produced by perfectly competitive firms combining a continuum of intermediate goods. Intermediate producers operate under monopolistic competition, use labor N_t with a linear production technology, and face TFP shocks a_t , also following an AR(1). Price changes are subject to adjustment costs, and we denote the resulting inflation by Π_t . The government maintains a balanced budget, and monetary policy follows a Taylor-type rule, responding to deviations of inflation and output from their steady-state values, subject to exogenous shocks m_t , which likewise follow an AR(1) process.

A minimal state-space representation of the model consists of the state variables $\mathbb{S} = \{a_t, d_t, m_t\}$ and policy functions $\{\Pi(\mathbb{S}), N(\mathbb{S})\}$. These are determined by three standard equilibrium conditions: the Euler equation (IS curve), the pricing condition (New Keynesian Phillips Curve), and the Taylor Rule. Given the model is highly standard, we leave most model details to Appendix C.

Model Solution We solve the model above using local linear approximations, that is, Taylor Projections (Levintal, 2018) of first-order. Given the minimal state representation stated above, we need to solve for two policy functions:

$$\begin{aligned} g_\Pi(a_t, d_t, m_t; x) &= \alpha_{0,x} + \alpha_{1,x}a_t + \alpha_{2,x}d_t + \alpha_{3,x}m_t \\ g_N(a_t, d_t, m_t; x) &= \beta_{0,x} + \beta_{1,x}a_t + \beta_{2,x}d_t + \beta_{3,x}m_t, \end{aligned}$$

where the coefficients $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \beta_0, \beta_1, \beta_2, \beta_3$ are allowed to differ at each point of the state-space where the local approximation is computed, reflected by subscript x . In Appendix C, we show how expectation operators of equilibrium conditions can be solved in closed form, resulting in large computational gains.

State-Space Representation Next, we illustrate how to map the model into state space form. The states of the model are given by $x_t = (a_t, d_t, m_t)$. The state-transition equation is not time-varying,

and follows:

$$\begin{pmatrix} a_{t+1} \\ d_{t+1} \\ m_{t+1} \end{pmatrix} = \begin{pmatrix} \rho_a & 0 & 0 \\ 0 & \rho_d & 0 \\ 0 & 0 & \rho_m \end{pmatrix} \begin{pmatrix} a_t \\ d_t \\ m_t \end{pmatrix} + \begin{pmatrix} \varepsilon_{t+1}^a \\ \varepsilon_{t+1}^d \\ \varepsilon_{t+1}^m \end{pmatrix}, \quad \begin{pmatrix} \varepsilon_t^a \\ \varepsilon_t^d \\ \varepsilon_t^m \end{pmatrix} \sim N(0, \text{diag}(\sigma_a^2, \sigma_d^2, \sigma_m^2))$$

Assume we observe log output y_t^{obs} , log inflation π_t^{obs} , and log interest rates r_t^{obs} . Our observation equation, conditional on having solved the model at state x , is given by:

$$\begin{pmatrix} y_t^{\text{obs}} \\ \pi_t^{\text{obs}} \\ r_t^{\text{obs}} \end{pmatrix} = \begin{pmatrix} \beta_{0,x} \\ (\pi_{ss} + \alpha_{0,x}) \\ (r_{ss} + \psi_\pi \alpha_{0,x} + \psi_y \beta_{0,x}) \end{pmatrix} + Z_x \begin{pmatrix} a_t \\ d_t \\ m_t \end{pmatrix} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \text{diag}(\sigma_y^{ME}, \sigma_\pi^{ME}, \sigma_r^{ME})^2)$$

$$\text{where } Z_x = \begin{pmatrix} \beta_{1,x} + 1 & \beta_{2,x} & \beta_{3,x} \\ \alpha_{1,x} & \alpha_{2,x} & \alpha_{3,x} \\ (\psi_\pi \alpha_{1,x} + \psi_y(\beta_{1,x} + 1)) & (\psi_\pi \alpha_{2,x} + \psi_y \beta_{2,x}) & (\psi_\pi \alpha_{3,x} + \psi_y \beta_{3,x} + 1) \end{pmatrix},$$

and σ_y^{ME} , σ_π^{ME} and σ_r^{ME} represent the measurement error standard deviations of log output, inflation and interest rates, respectively. Note that the bottom row in Z reflects how interest rates depend on both output and inflation through the monetary policy rule. This linear state-space representation allows us to solve and filter the model as in Algorithm 1.

Estimation details For estimation, we use logged real GDP-per-capita.¹⁴ Inflation is quarterly inflation based on the consumer price index¹⁵ and the nominal rate is the quarterly equivalent of the annual federal funds rate¹⁶. We estimate the model on the 1966:Q1-2007:Q4 sample. We use a Random-Walk Metropolis Hastings sampler, with the proposal variance of the algorithm set to the variance-covariance matrix of the loglinearized posterior draws, scaled to obtain an acceptance rate of 30%. We use the same priors as for the linear model, presented in Table 2. We calibrate a subset of parameters: markup $\gamma = 0.1$, and inverse Frisch $\eta = 5$.

Runtime In Matlab, on a standard laptop without parallelization, it takes 7 minutes and 50 seconds to get 100,000 draws from the nonlinear model, about 0.0047 seconds per draw.¹⁷ This is less than one-and-a-half times longer than it takes to estimate the linear model in Dynare (Adjemian et al., 2024), which takes five and a half minutes. For comparison, for a similar model, An and Schorfheide (2007) indicate that particle filtering with second-order perturbations is about 200 times slower than using a Kalman filter with first-order perturbation. A global model solution is

¹⁴FRED: GDPC1 divided by FRED: CNP16OV, detrended using the HP-filter with a smoothing parameter of 1,600.

¹⁵FRED: CPIAUCSL

¹⁶FRED: DFF

¹⁷Computations are done on a Apple Macbook Pro with an Apple M3 Max processor and 64GB RAM.

at least an order of 500 times slower than a perturbation solution, even for models with a smaller state space like this one (Kollmann et al., 2011).

Comparing Linear and Nonlinear Estimates Table 2 report the priors – which are common across both the linear model and nonlinear model – and the resulting posteriors for both the linear and nonlinear model estimates. The nonlinear model has lower measurement error standard deviations, suggesting better fit. For a subset of parameters, the linear and nonlinear model agree closely on the shape and location of the posterior distribution. The biggest differences are in the price adjustment cost parameter ϕ_p which increases from 49 to 76 when estimating the model nonlinearly, and the estimates of the monetary policy shock process, where the nonlinear model indicates lower persistence of shocks, but with a larger variance.

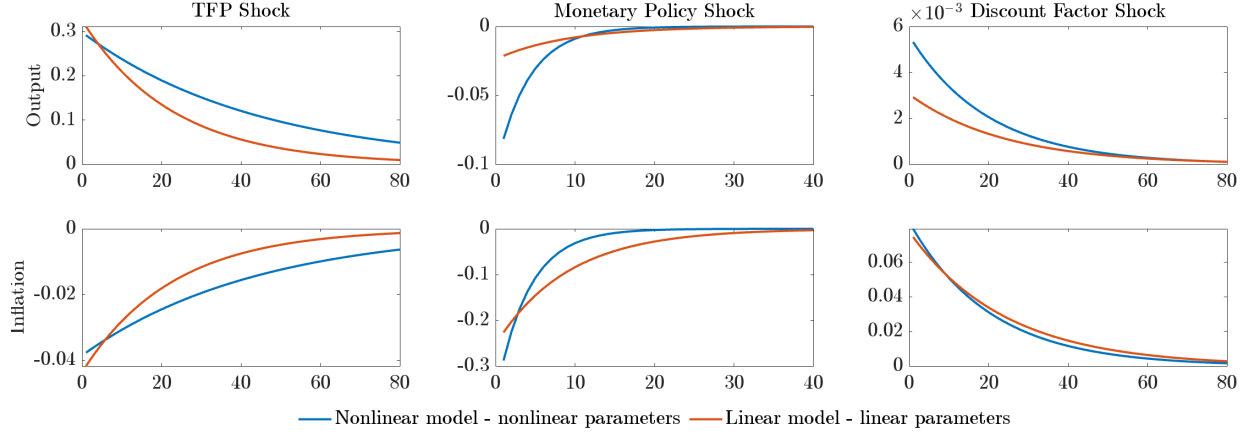
Table 2: Prior mean and standard deviation, posterior mean and 90% CI

Parameter	Description	Prior (Mean, Std.)	Linear Posterior Mean	Linear Posterior [5%, 95%]	Nonlinear Posterior Mean	Nonlinear Posterior [5%, 95%]
Economic Parameters						
τ	Inverse elasticity of substitution	G (2, 0.5)*	1.51	[1.22, 1.80]	1.68	[1.07, 2.55]
ϕ_p	Price adjustment cost	G(50, 10)*	49.3	[43.5, 56.7]	76.0	[51.5, 101]
ψ_y	Taylor rule output	G (0.75, 0.25)	0.18	[0.09, 0.27]	0.18	[0.08, 0.30]
ψ_π	Taylor rule inflation	G (2.0, 0.25)	2.56	[2.36, 2.92]	2.63	[2.26, 3.06]
π_{ss}	Steady state inflation	G (0.01, 0.005)	0.012	[0.010, 0.013]	0.011	[0.008, 0.013]
r_{ss}	Steady state interest rate	G (0.016, 0.005)	0.016	[0.015, 0.018]	0.014	[0.009, 0.019]
Exogenous processes						
ρ_m	Persistence monetary shock	B (0.7, 0.15)	0.88	[0.80, 0.94]	0.78	[0.68, 0.91]
ρ_d	Persistence discount factor shock	B (0.7, 0.15)	0.96	[0.94, 0.97]	0.95	[0.92, 0.98]
ρ_a	Persistence TFP shock	B (0.7, 0.15)	0.96	[0.92, 0.99]	0.98	[0.95, 1.00]
$100 \times \sigma_m$	Standard dev. monetary shock	IG (0.5, 1)	0.40	[0.31, 0.52]	0.58	[0.34, 0.82]
$10 \times \sigma_d$	Standard dev. discount factor shock	IG (0.5, 0.5)	0.28	[0.19, 0.45]	0.28	[0.16, 0.48]
$100 \times \sigma_a$	Standard dev. TFP shock	IG (1, 1)	0.34	[0.30, 0.38]	0.33	[0.27, 0.39]
Measurement error						
$100 \times \sigma_y^{ME}$	ME output	IG(0.5, 200)	0.12	[0.09, 0.15]	0.08	[0.05, 0.12]
$100 \times \sigma_\pi^{ME}$	ME inflation	IG(0.5, 200)	0.43	[0.38, 0.48]	0.41	[0.37, 0.47]
$100 \times \sigma_r^{ME}$	ME interest rate	IG(0.5, 200)	0.10	[0.08, 0.12]	0.06	[0.04, 0.08]
Computation time						
Time			5 min 30 s		7 min 50 s	

Note: G is Gamma, IG is inverse Gamma, B is Beta. τ 's prior is truncated at 1 from below, and ϕ_p 's prior is truncated at 200 from above.

Figure 3 compares the impulse response functions of the linear and nonlinear model, evaluated at the steady state, which show the implications of different model solutions and parameter estimates. In the nonlinear model estimates, the real effects of monetary policy are larger, evidencing a flatter Phillips curve than implied by the linear model.

Figure 3: Impulse Response Functions of Estimated Linear and Nonlinear Model



4 Main Application: A New Keynesian Model with Labor Search

In this section, we present a New Keynesian (NK) model with labor market frictions, following the Diamond-Mortensen-Pissarides (DMP) framework. The model illustrates how the strength of monetary policy transmission depends on firms' flow surplus, leading to state-dependent effects of monetary policy. We provide intuition for this result by analytically log-linearizing the model, showing that surplus scales the sensitivity to shocks. We show how to solve the model using local linear solution methods, how to map the model solution in time-varying state-space form, and consequently how to estimate the model. Using the estimated model and filtered states, we provide both model-based and reduced-form evidence that the Phillips Multiplier, that is, the inflation-unemployment trade-off, is time-varying and decreases during recessions.

4.1 Model

Overview The economy consists of households that consume and supply labor, intermediate good producers that set prices and hire labor in a frictional labor market where workers seek to match with jobs, and final good producers that aggregate intermediate goods while taking prices as given. Wages are determined through Nash bargaining between workers and intermediate goods firms. The government sets the nominal interest rate according to an inertial Taylor rule.

Timing We follow the same timing assumption as the DMP model in Section 3. Each model period consists of four sub-periods: (i) separations, (ii) search and matching, (iii) bargaining, and (iv) production and consumption/saving.

Households Households are composed of a continuum of members $i \in [0, 1]$ who derive utility from consumption $c_t(i)$, and supply one unit of labor $n_t(i) \in \{0, 1\}$ if employed and none if unemployed. The members of the household perfectly insure each other against variations in

income due to unemployment. Hence, defining $C_t := \int c_t(i)di$, and $N_t := \int n_t(i)di$, households maximize:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t D_t \frac{C_t^{1-\tau} - 1}{1-\tau} \right],$$

subject to the budget constraint and law of motion for employment:

$$\begin{aligned} P_t C_t + B_t &= P_t W_t N_t + P_t (1 - N_t) b + R_{t-1} B_{t-1} + T_t, \\ N_t &= (1 - \delta_t) N_{t-1} + q_t V_t. \end{aligned}$$

Here β is the discount factor, D_t is a shock to the value of consumption, $1/\tau$ is the intertemporal elasticity of substitution, P_t is the price of the final good, B_t denotes one-period risk-free bonds with gross return R_t , and T_t denotes lump sum taxes net of profits transferred from firms. W_t are real wages, and b denotes the resources provided by the unemployed (e.g., unemployment benefits and home production). The law of motion for employment depends on the exogenous separation rate δ_t and the number of new hires $q_t V_t$ made by firms which the household takes as given, with V_t the number of vacancies and q_t the vacancy filling rate.

The first-order condition with respect to consumption implies

$$\lambda_t = D_t C_t^{-\tau},$$

where λ_t is the marginal utility of consumption. Defining the stochastic discount factor used by firms as $Q_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}$, the first-order conditions with respect to bonds and consumption imply

$$1 = \beta E_t \left[R_t \frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau} \frac{P_t}{P_{t+1}} \right] = E_t \left[Q_{t,t+1} \frac{R_t}{\Pi_{t+1}} \right]$$

where $\Pi_t = P_t/P_{t-1}$ is inflation. Defining the surplus to the household of employment S_t^h using the envelope condition, we obtain

$$S_t^h = W_t - b + E_t [Q_{t,t+1} (1 - \delta_{t+1}) S_{t+1}^h].$$

which denotes the present value of a marginal employed household member.

Search and Matching The number of aggregate hires is determined by a constant returns to scale matching function $M_t = A_t M(V_t, S_t)$ where A_t is aggregate matching efficiency, V_t is aggregate vacancies and S_t is the number of unmatched workers searching for jobs. Given the timing assumptions, the mass of unmatched workers in the search stage is given by

$$S_t = 1 - N_{t-1} + \delta_t N_{t-1} = 1 - (1 - \delta_t) N_{t-1},$$

that is, the workers who were not matched at the end of the previous production stage plus those who incurred an involuntary separation at the beginning of the period. The probability a vacancy contacts workers can be summarized by the market tightness $\theta_t = V_t/S_t$ as $q_t = M_t/V_t = A_t M(1, \theta_t^{-1})$.

Final Goods Producers The final good producer aggregates intermediate goods $Y_t(j)$, $j \in [0, 1]$, using the technology:

$$Y_t = \left(\int_0^1 Y_t(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}}.$$

The representative firm maximizes profits taking input prices $P_t(j)$ and output prices P_t as given. Profit maximization implies that the demand for input j is given by:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\frac{1}{\gamma}} Y_t.$$

Intermediate Goods Producers There is a continuum $j \in [0, 1]$ of identical monopolistic intermediate goods producers, with production technology

$$Y_t(j) = Z_t N_t(j)$$

where Z_t is a neutral technology shock, and $N_t(j)$ is the quantity of labor. These firms are monopolists in the product market and hire in frictional labor markets. Price adjustments are subject to a Rotemberg (1982) quadratic adjustment cost

$$AC_t(j) = \frac{\phi}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right)^2 Y_t,$$

where ϕ is an adjustment cost parameter and Π is steady state inflation. Firms hire labor in a frictional labor market by posting vacancies at a fixed cost c . Hence, the firm's problem is to choose employment $N_t(j)$, vacancies $V_t(j)$, and prices $P_t(j)$ to maximize the present value of real profits discounted by $Q_{0,t}$:

$$\Pi_0(j) = \mathbb{E}_0 \sum_{t=0}^{\infty} Q_{0,t} \left\{ \left(\frac{P_t(j)}{P_t} Z_t - W_t \right) N_t(j) - c V_t(j) - AC_t(j) \right\}.$$

subject to

$$Z_t N_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\frac{1}{\gamma}} Y_t, \quad [\omega_t]$$

$$N_t(j) = (1 - \delta_t) N_{t-1}(j) + q_t V_t(j). \quad [\mu_t]$$

where the first constraint implies that the choice of output $Y_t(j) = Z_t N_t(j)$ and prices $P_t(j)$ are constrained to lie on the firms' product demand curves. The multiplier on this constraint is given by ω_t . This leads to both an explicit cost of labor the real wage W_t and an implicit cost ω_t as hiring a marginal worker – all else equal – leads to a reduction in prices as firms move down their residual demand curve. The second constraint is the law of motion for employment at the firm and the multiplier μ_t represents the value of a marginal worker to the firm.

Wage Bargaining The wage is determined by the Nash sharing rule

$$\eta \mu_t = (1 - \eta) S_t^h.$$

where η is the workers' bargaining weight. This leads to the following expression for the real wage

$$W_t = \eta(1 - \omega_t)Z_t + (1 - \eta)b$$

where b is the flow value of non-employment and ω_t is the implicit marginal cost of hiring.

Government Monetary policy is determined by a Taylor Rule for the nominal interest rate subject to a ZLB constraint:

$$R_t = \max \{1, \tilde{R}_t e^{m_t}\},$$

where $\tilde{R}_t$ is the systematic component of monetary policy that reacts to the state of the economy, and m_t represents an inertial deviation from the deterministic Taylor Rule. We consider the following Taylor rule:

$$\tilde{R}_t = R \left(\frac{\Pi_t}{\Pi} \right)^{\psi_\pi} \left(\frac{Y_t}{Y} \right)^{\psi_y},$$

where $\Pi_t = P_t/P_{t-1}$ is inflation, and Y_t is output. Note that the nominal interest rate is almost everywhere differentiable, except at the kink where the ZLB begins to bind. The local perturbation method we employ assumes that the equilibrium conditions of the model are differentiable in the state variables. Hence, to accommodate the ZLB, we smooth the kink using the softmax function

$$R_t = \zeta \log \left(e^{1/\zeta} + e^{\tilde{R}_t e^{m_t}/\zeta} \right),$$

where the parameter ζ determines how smooth the approximation is and has the property

$$\lim_{\zeta \rightarrow 0} R_t = \max \{1, \tilde{R}_t e^{m_t}\}.$$

The government levies lump-sum taxes (or provides a subsidy) to run a balanced budget every period.

Exogenous States The discount factor $d_t = \ln D_t$, TFP $z_t = \ln Z_t$, and the monetary shock m_t are assumed to be AR(1) in logs. The separation rate $\delta_t = \delta e^{s_t}$ and matching efficiency $A_t = A e^{a_t}$ are assumed to have stochastic components that are AR(1) in logs as well.

$$\begin{aligned} z_t &= \rho_z z_{t-1} + \sigma_z \epsilon_t^z \\ d_t &= \rho_d d_{t-1} + \sigma_d \epsilon_t^d \\ m_t &= \rho_m m_{t-1} + \sigma_m \epsilon_t^m \\ s_t &= \rho_s s_{t-1} + \sigma_s \epsilon_t^s \\ a_t &= \rho_a a_{t-1} + \sigma_a \epsilon_t^a \end{aligned}$$

Equilibrium Conditions We assume vacancy costs and adjustment costs are paid in lump-sum fashion to households, so market clearing implies

$$C_t = Y_t = Z_t N_t.$$

This leads to the following Euler equation

$$1 = \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right].$$

In a symmetric equilibrium all firms set the same prices $P_t = P_t(j)$ and choose the same level of employment $N_t(j) = N_t$ and vacancies $V_t(j) = V_t$ leading to the following pricing equation

$$\frac{\omega_t}{\gamma} = 1 - \phi(\Pi_t - \Pi)\Pi_t + \phi\beta\mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{1-\tau} (\Pi_{t+1} - \Pi)\Pi_{t+1} \right].$$

Similarly, optimal vacancy posting and the expression for the Nash wage leads to the following job-creation condition

$$\frac{c}{q_t} = (1 - \eta)[(1 - \omega_t)Z_t - b] + \beta\mathbb{E}_t \left[(1 - \delta_{t+1}) \frac{D_{t+1}}{D_t} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\tau} \frac{c}{q_{t+1}} \right]$$

The Taylor Rule for the nominal rate and the law of motion for employment close the model. A minimum state variable representation of the model has six states $\mathbb{S}_t = \{N_{t-1}, z_t, d_t, m_t, s_t, a_t\}$ and three policies $\{\Pi(\mathbb{S}_t), \omega(\mathbb{S}_t), \theta(\mathbb{S}_t)\}$.

4.2 State-Dependence

We use the system of log-linearized conditions to think about the deterministic response of the system to a one-time monetary shock $\epsilon_0^m = 1$, and $\epsilon_t^m = 0 \quad \forall t > 0$, that is, the perfect foresight path. Omitting expectation operators as this is the perfect foresight response, and focusing on two

linearized equations in particular, we have:

$$\begin{aligned} \text{Pricing/NK-Phillips: } \hat{\pi}_t &= \beta \hat{\pi}_{t+1} - \frac{1}{\phi} \hat{\omega}_t \\ \text{Job-Creation: } \hat{q}_t &= \frac{\gamma(1 - \beta(1 - \delta))}{(1 - \eta)(1 - \gamma - b)} \hat{\omega}_t + (1 - \delta)\beta [\hat{q}_{t+1} + \hat{r}_t - \hat{\pi}_{t+1}] \end{aligned}$$

As evident from the log-linearized job creation condition, the sensitivity of q_t to ω_t scales inversely with $(1 - \eta)(1 - \gamma - b)$, which represents the steady-state flow profit of the firm. This is closely related to the *fundamental surplus* concept in Ljungqvist and Sargent (2017). When the flow profit is small, shocks propagate more strongly to the vacancy-filling rate q_t , and vice versa. Since q_t determines market tightness, the same mechanism governs the propagation of shocks to market tightness and, consequently, employment and output. Given that inflation depends on $\hat{\omega}_t$ through the New Keynesian Phillips curve, this mechanism also drives state dependence in inflation. The strength of inflation's response to a monetary shock is influenced by the degree of price stickiness ϕ , as specified in the Rotemberg (1982) adjustment cost framework.

In a log-linearized NK-DMP model, the propagation mechanisms depend on steady-state profit $(1 - \eta)(1 - \gamma - b)$ for real variables and on price stickiness ϕ for inflation. However, these elasticities remain constant across different economic states. In a nonlinear solution, flow profit and hence shock propagation vary across the state space, resulting in state-dependent monetary policy transmission, which is the focus of Section 5.

4.3 Solution and Estimation

Here we describe the model solution and mapping to observables in the state space model. First, we define states and policies in terms of log-deviations from their steady state values, e.g. $\theta_t = \ln(\Theta_t/\Theta)$ and $z_t = \ln(Z_t/Z)$. Next, we define the equilibrium conditions in terms of these policies under the assumption that employment n_t , inflation π_t , the marginal cost ω_t and market tightness are linear in the states $x_t = \{n_{t-1}, z_t, d_t, m_t, s_t, a_t\}$. This gives policies of the form.

$$\begin{aligned} \theta_t &= b_{0,x}^\theta + B_x^\theta x_t \\ \pi_t &= b_{0,x}^\pi + B_x^\pi x_t \\ \omega_t &= b_{0,x}^\omega + B_x^\omega x_t \\ n_t &= b_{0,x}^n + B_x^n x_t, \end{aligned}$$

where each B_x^i term is a 1×6 vector of coefficients. Using these local linear policies, the expectation terms of the equilibrium system can be solved in closed form, and the coefficients can be recovered using the Taylor Projection method described in Section 2. The softmax function used for the ZLB introduces an additional nonlinear equation. To maintain a linear state-space representation, we

approximate the interest rate using a first-order Taylor expansion.

$$r_t = b_{0,x}^r + B_x^r x_t.$$

At states where the interest rate is far from the ZLB, the linear approximation closely matches the linear unconstrained form $r_t = \psi_1 \pi_t + \psi_2 (n_t + z_t) + m_t$. However, as the interest rate approaches the ZLB, the dynamics differ. Additional details and derivations of the local linear solution are provided in Appendix D.

With the local solution to the equilibrium system in hand, we define the state-space mapping between observables and latent states. We consider five observable variables: market tightness $\tilde{\theta}_t$, job-finding rate $\tilde{f}_t$, employment $\tilde{n}_t$, Federal Funds rate $\tilde{r}_t$, and inflation $\tilde{\pi}_t$, where the tildes indicate that these observables may be measured with error. The mapping between these observables and the model variables incorporates measurement errors and is given by:

$$\begin{aligned}\tilde{\pi}_t &= \pi + \pi_t + \sigma_\pi^{me} \epsilon_t^\pi \\ \tilde{n}_t &= n_t + \sigma_n^{me} \epsilon_t^n \\ \tilde{\theta}_t &= \theta_t + \sigma_\theta^{me} \epsilon_t^\theta \\ \tilde{r}_t &= r + r_t + \sigma_r^{me} \epsilon_t^r \\ \tilde{f}_t &= f_t + \sigma_f^{me} \epsilon_t^f = a_t + (1 - \alpha)\theta_t + \sigma_f^{me} \epsilon_t^f\end{aligned}$$

where π is the natural log of steady-state inflation and r is the log of the steady-state nominal rate. Letting $y_t^{\text{obs}} = (\tilde{\pi}_t, \tilde{n}_t, \tilde{\theta}_t, \tilde{r}_t, \tilde{f}_t)'$, the state space for of the model at time t is given by

$$\begin{aligned}y_t^{\text{obs}} &= d_t + Z_t x_t + \epsilon_t^{me}, \quad \epsilon_t^{me} \sim N(0, Q) \\ x_t &= c_t + T_t x_{t-1} + R \eta_t^x, \quad \eta_t^x \sim N(0, H)\end{aligned}$$

where the measurement equation matrices are

$$d_t = \begin{pmatrix} \pi + b_{0,t}^\pi \\ 0 \\ b_{0,t}^\theta \\ r + b_{0,t}^r \\ (1 - \alpha)b_{0,t}^\theta \end{pmatrix} \quad Z_t = \begin{pmatrix} b_{1,t}^\pi & b_{2,t}^\pi & b_{3,t}^\pi & b_{4,t}^\pi & b_{5,t}^\pi & b_{6,t}^\pi \\ 1 & 0 & 0 & 0 & 0 & 0 \\ b_{1,t}^\theta & b_{2,t}^\theta & b_{3,t}^\theta & b_{4,t}^\theta & b_{5,t}^\theta & b_{6,t}^\theta \\ b_{1,t}^r & b_{2,t}^r & b_{3,t}^r & b_{4,t}^r & b_{5,t}^r & b_{6,t}^r \\ (1 - \alpha)b_{1,t}^\theta & (1 - \alpha)b_{2,t}^\theta & (1 - \alpha)b_{3,t}^\theta & (1 - \alpha)b_{4,t}^\theta & (1 - \alpha)b_{5,t}^\theta & (1 - \alpha)b_{6,t}^\theta + 1 \end{pmatrix}$$

and the state equation matrices are

$$c_t = \begin{pmatrix} b_{0,t}^n \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad T_t = \begin{pmatrix} b_{1,t}^n & b_{2,t}^n & b_{3,t}^n & b_{4,t}^n & b_{5,t}^n & b_{6,t}^n \\ 0 & \rho_z & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho_d & 0 & 0 & 0 \\ 0 & 0 & 0 & \rho_m & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_s & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_a \end{pmatrix}$$

We estimate the model on the sample period from 1966:Q1-2019:Q4. The data used for estimation are summarized in Table B1 in the Appendix. This table also details the data transformations applied.

4.4 Details on Empirical Framework: State-Dependent Local Projections

In addition to the model based results, in the next section, we also provide empirical model-free evidence of state-dependence in monetary transmission. Therefore, we briefly discuss the empirical framework we use for this. Following Cloyne et al. (2023), one can extend the standard Local Projection (Jordà, 2005) framework to allow for state-dependent impulse responses. Specifically, consider a *state variable* x_t . The state-dependent local projection framework is given by:

$$y_{t+h} = \alpha_h + \beta_h s_t + \beta_h^x s_t (x_t - \bar{x}) + w_t \gamma_h + \varepsilon_t, \quad \text{for } h = 0, \dots, H \text{ and } t = 1, \dots, T, \quad (28)$$

where w_t is a set of control variables, augmented with $(x_t - \bar{x})$, representing deviations of the state variable from its steady-state level $\bar{x}$. Relative to a standard Local Projection, the state-dependent specification introduces the term $\beta_h^x s_t (x_t - \bar{x})$, which captures the interaction between the shock s_t and the state variable x_t . The state-dependent impulse response function is then given by the sum of the direct effect of the shock, $\hat{\beta}_h$, and the state-dependent effect, $\hat{\beta}_h^x (x_t - \bar{x})$.¹⁸ We use an instrumental variable (IV) approach, where the shock s_t is replaced by a policy variable, such as the Federal Funds Rate (FFR), instrumented in a first stage using an exogenous shock series. Given the model includes an interaction term, the instrument set is extended by the interaction between the exogenous shock and the deviation of the state variable from its steady-state level.

A causal interpretation of this framework relies on the exogeneity of the monetary policy shocks with respect to both state and outcome variables at all horizons (Gonçalves, Herrera, Kilian, and Pesavento, 2023) — a condition that may not be satisfied. Our aim is more modest: to document correlations indicative of state dependence that are consistent with the theoretical model.

Outcome Variables The empirical analysis focuses on the effects of monetary policy shocks on the unemployment rate, vacancies (from the Job Openings and Labor Turnover Survey, and measured

¹⁸To aid inference, we impose the restriction that the response for each month in a quarter is the same, reducing the number of parameters to be estimated and shortening confidence intervals.

using the Help-Wanted Index from Barnichon (2010) before December 2001), market tightness (the ratio of vacancies to unemployment) and the Consumer Price Index (CPI). Detailed descriptions of the data sources and transformations are provided in Appendix B.

Controls We follow Ramey (2016) by including controls for the CPI, the unemployment rate, the Producer Price Index (PPI), and lags of the monetary policy variable and the dependent variable. Following the recommendations of Bauer and Swanson (2023), the LP-IV framework includes 12 monthly lags, and we add the excess bond premium as an additional control.

State Proxies The empirical analysis in this section uses two proxies likely correlated with firm surplus to provide model-free reduced-form evidence on state-dependent monetary policy transmission. First, we consider Corporate Profits. The value of job creation corresponds to the present value of profits obtained from a marginal hire. While this present value is not directly observable in the data, corporate profits are. In the data, corporate profits are measured as the ratio of profits to the sum of profits and compensation, as reported in the National Income and Product Accounts (NIPA), Table 1.10, filtered using the Hamilton filter (Hamilton, 2018). A second proxy is the Unemployment Rate: this is consistent with model predictions which would indicate that unemployment tracks flow surplus, albeit with a potential lag. We again first apply the Hamilton filter (Hamilton, 2018).

Instruments and Policy Variables We employ three commonly used instruments for the monetary policy rate (which we replace by the one-year U.S. Treasury yield): (i) the narrative monetary policy shock series from Romer and Romer (2004), extended by Wieland and Yang (2020) to 2007; (ii) the language-processing approach of Aruoba and Drechsel (2024), spanning 1982-2008; (iii) the high-frequency approach of Bauer and Swanson (2023), spanning 1988-2019.

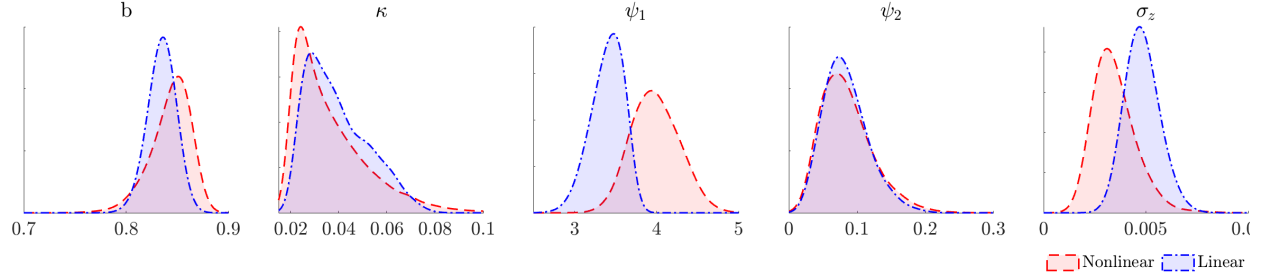
5 Model Results

We now present results from the quantitative model, solved using Taylor projections and using the Time-Varying Kalman Filter to infer the underlying states of the economy. This approach enables us to compute implied impulse response functions over time, as well as other objects of interest to policymakers, and to quantify state-dependence in monetary policy transmission.

5.1 Parameter Estimates

A subset of parameters is calibrated, as reported in Table 3. Priors and posteriors are summarized in Table 4, comparing estimates from the linear and nonlinear models. It takes about 2.5 hours to estimate the nonlinear model. A selected subset of these posteriors is shown in Figure 4. The estimates differ in several respects. First, the standard deviation of total factor productivity (TFP) shocks is larger in the linear model, reflecting weaker endogenous propagation. Second, the Taylor

Figure 4: Selected Posteriors: linear solution with Kalman filter versus and nonlinear solution (Taylor Projection) with Time-Varying Kalman Filter



rule places greater weight on inflation in the nonlinear model. The linear model is estimated on a sample ending in 2007 to avoid the period when the zero lower bound (ZLB) binds, which may further contribute to the differences.

Note that the outside option of workers, b , determines the size of the surplus in the model. To rationalize observed unemployment fluctuations, b must be large relative to the joint surplus of workers and firms. We estimate b by introducing an auxiliary variable $\nu \in [0, 1]$, where

$$b = \nu(1 - \gamma)e^{-3\sigma_z/\sqrt{1-\rho_z^2}}.$$

This construction ensures positive surplus and vacancies across draws of σ_z and ρ_z . Although the posterior of ν is comparable across both model solutions, the implied distribution of b varies. Figure 4 shows selected posteriors. Differences in estimation and solution methods also affect the filtered states: as shown in Figure 5, TFP and discount factor shocks are more volatile under the linear model, in line with parameter estimates. .

Table 3: Calibrated parameters

Parameter	Calibrated value
η : bargaining weight	0.5
τ : risk aversion	2
γ : inverse elasticity of substitution CES aggregator	0.1
q_{ss} : steady state vacancy-filling rate	0.7
θ_{ss} : steady state market tightness	1
n_{ss} : steady state employment	0.945
δ : exogenous job destruction rate	$\frac{q_{ss}\theta_{ss}(1-n_{ss})}{(1-q_{ss}\theta_{ss})n_{ss}}$
s_{ss} : steady state separation rate	$\log \delta$
a_{ss} : steady state match efficiency	$\log q_{ss}$
c : vacancy cost	$\frac{(1-\eta)(1-\gamma-b)q_{ss}}{1-\beta(1-\delta)}$

Table 4: Prior distribution with mean and standard deviation, posterior means and standard deviations for log-linear and TP+TV-KF (based on 100,000 draws, acceptance rate of $\approx 34\%$, burn-in 20,000, thin 3).

Parameter	Prior Distr. (Mean, Std.)	Posterior			
		Log-linear		TP TV-KF	
		Mean	Std.	Mean	Std.
Economic parameters					
ν : flow value unemployment	Beta(0.9, 0.025)*	0.971	0.006	0.976	0.006
κ : slope Phillips curve	Beta(0.05, 0.02)*	0.039	0.013	0.037	0.015
α : matching function	Beta(0.5, 0.1)	0.816	0.023	0.837	0.025
ψ_1 : inflation Taylor rule	Gamma(1.5, 0.25)	3.401	0.181	3.985	0.281
ψ_2 : output Taylor rule	Gamma(0.25, 0.1)	0.084	0.032	0.085	0.037
Exogenous process parameters					
σ_z : std. TFP shocks	InvGamma(0.02, 0.05)	0.005	0.001	0.004	0.001
σ_d : std. DF shocks	InvGamma(0.05, 0.1)	0.017	0.003	0.013	0.002
σ_s : std. separation shocks	InvGamma(0.05, 0.1)	0.021	0.004	0.042	0.004
σ_m : std. MP shocks	InvGamma(0.02, 0.05)	0.005	0.001	0.005	0.001
σ_a : std. matching efficiency shocks	InvGamma(0.02, 0.05)	0.011	0.001	0.012	0.001
ρ_z : persistence TFP	Beta(0.5, 0.1)	0.949	0.004	0.964	0.010
ρ_d : persistence DF	Beta(0.5, 0.1)	0.890	0.020	0.874	0.018
ρ_s : persistence separations	Beta(0.5, 0.1)	0.770	0.059	0.712	0.038
ρ_r : inertia Taylor rule	Beta(0.5, 0.1)	0.883	0.025	0.886	0.023
ρ_a : persistence matching efficiency	Beta(0.5, 0.1)	0.885	0.028	0.909	0.020
Measurement error					
σ_π^{ME} : ME inflation	InvGamma(0.005, 0.05)	0.0018	0.0002	0.0018	0.0001
σ_N^{ME} : ME employment	InvGamma(0.005, 0.05)	0.0008	0.0001	0.0006	0.0001
σ_θ^{ME} : ME tightness	InvGamma(0.015, 0.05)	0.0152	0.0057	0.0368	0.0045
σ_r^{ME} : ME FFR	InvGamma(0.005, 0.05)	0.0010	0.0001	0.0006	0.0001
σ_s^{ME} : ME separations	InvGamma(0.015, 0.05)	0.0158	0.0018	0.0179	0.0015
Steady states					
π^{ss} : steady state inflation	InvGamma(0.008, 0.02)	0.0087	0.0015	0.0097	0.0010
r_s : steady state interest rate	InvGamma(0.013, 0.02)	0.0129	0.0022	0.0138	0.0014

Note: (*) The prior for ν is truncated from above at 0.985 and for κ truncated from below at 0.02.

5.2 Time-Varying Impulse Responses

An attractive feature of the methodology proposed in this paper is that it allows for computing the implied impulse response functions at each point in time conditional on the filtered states. The extent to which shock propagation varies over time can be understood through the local linear policy coefficients B_x^θ , B_x^π , B_x^ω and B_x^n . In a truly linear model, these coefficients are constant, and impulse responses are invariant to the state. In our setting, this is not the case, as seen by Figure 6, which plots a subset of filtered policy coefficients, along with their 68% credible intervals reflecting parameter uncertainty. The policy coefficients can be interpreted as elasticities since all variables are expressed in log-deviations from steady state values. The results show that the coefficients vary

Figure 5: Filtered States for Linear and Nonlinear Model Solution

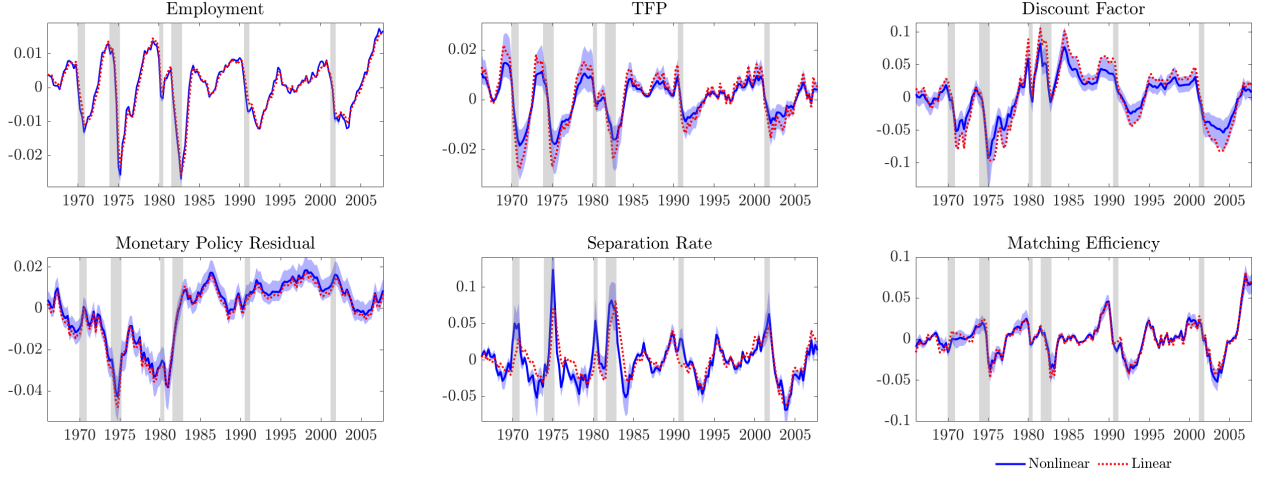
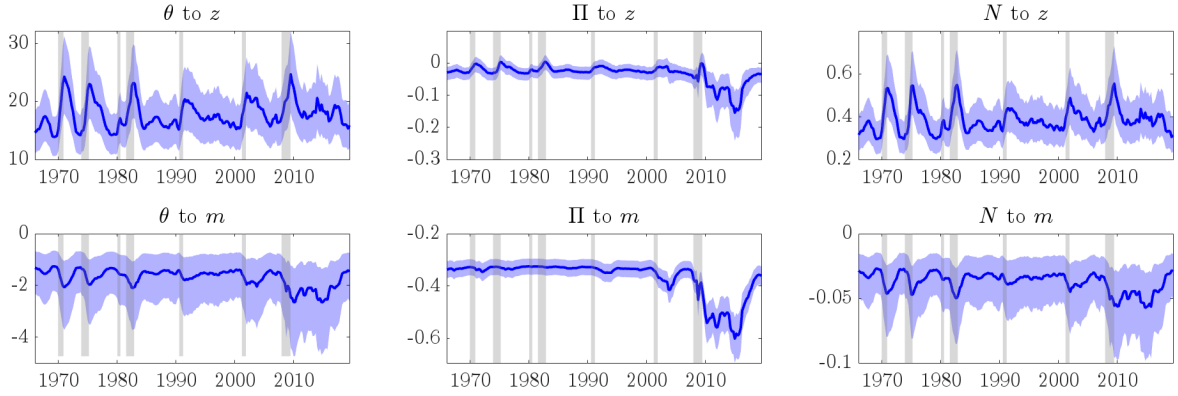


Figure 6: State-Dependent Policy Coefficients, computed at filtered states



systematically with the state. Elasticities are typically larger in absolute value during recessions, when the model-implied surplus is low, consistent with theoretical predictions. For instance, the effect of monetary shocks on market tightness is larger when unemployment is high, and once the ZLB binds, the effects of non-monetary shocks on inflation become large as monetary policy becomes constrained.

To further illustrate the extent of time variation, Figure 7 shows the impulse response functions for unemployment, market tightness, and vacancies at two different dates, computed based on the filtered states. The first date, 1973Q2, corresponds to a high-surplus period in which monetary policy shocks propagate little. The second date, 2008Q2, represents a low-surplus period with stronger propagation. In addition to the model-implied impulse responses, we also report results from three different state-dependent local projections, introduced in Section 4.4. Appendix E shows confidence bands for the baseline estimate and the interaction terms. The model's predictions for unemployment responses are broadly consistent with the local projections, both in magnitude and

amount of state dependence. While the model does not generate hump-shaped responses, which would require introducing habit formation or additional frictions, this limitation is not central to the analysis. The data show somewhat stronger responses in market tightness and vacancies than the model predicts, but the degree of state dependence is comparable.

Figure 7: *Impulse Response Functions over Time: 1973Q2 (blue dash-dot) and 2008Q2 (red solid)*



5.3 Phillips Multiplier

A key concern for policymakers is the unemployment-inflation trade-off, which captures the nominal effects relative to the real effects of a monetary shock. However, state dependence in impulse responses alone does not necessarily imply variation in the unemployment-inflation trade-off; this depends on whether the real versus nominal effects differ in relative magnitude across states. To illustrate that the unemployment-inflation trade-off is state dependent, we use the concept of the Phillips multiplier, as in Barnichon and Mesters (2021), motivated by Mankiw (2001). The Phillips multiplier quantifies this trade-off by measuring the average change in inflation caused

by an interest rate change that reduces the unemployment rate by one percentage point over h periods:

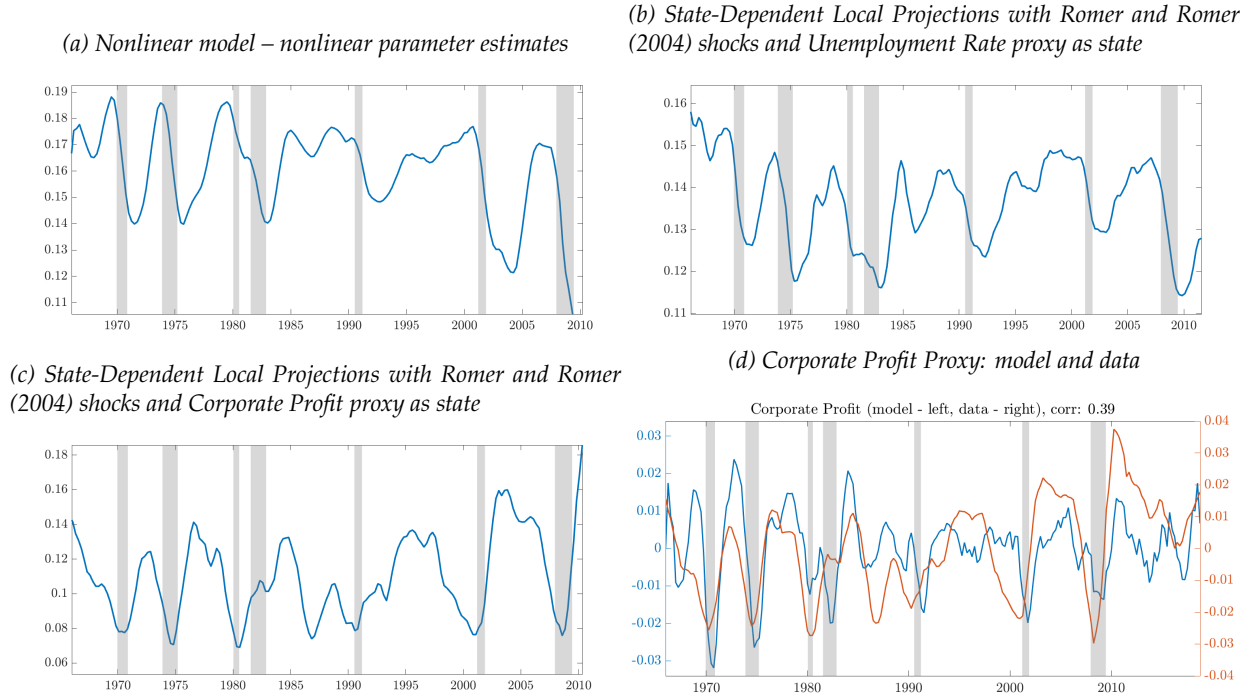
$$T_h = \frac{\partial \bar{\Pi}_{t:t+h}}{\partial R_t} \Big|_{\varepsilon_t} \Big/ \frac{\partial \bar{U}_{t:t+h}}{\partial R_t} \Big|_{\varepsilon_t}$$

Here $\bar{\Pi}_{t:t+h}$ is the average over variable Π over h time periods, and ε_t is the exogenous change in the interest rates. The statistical counterpart of T_h is the so-called Phillips multiplier (Barnichon and Mesters, 2021):

$$\mathcal{P}_h = \mathcal{R}_h^{\bar{\Pi}} / \mathcal{R}_h^{\bar{U}}, \quad h = 0, 1, 2, \dots,$$

that is, the ratio of the cumulative impulse response function (up to horizon h) of inflation over unemployment. The slope of the Phillips curve and the Phillips Multiplier are strongly connected, but not one-to-one. In the case of state-dependent impulse response functions, the Phillips Multiplier is also state-dependent when the magnitude of the effect of a monetary shock on inflation relative to unemployment depends on the state of the economy.

Figure 8: State-Dependence in the Phillips Multiplier



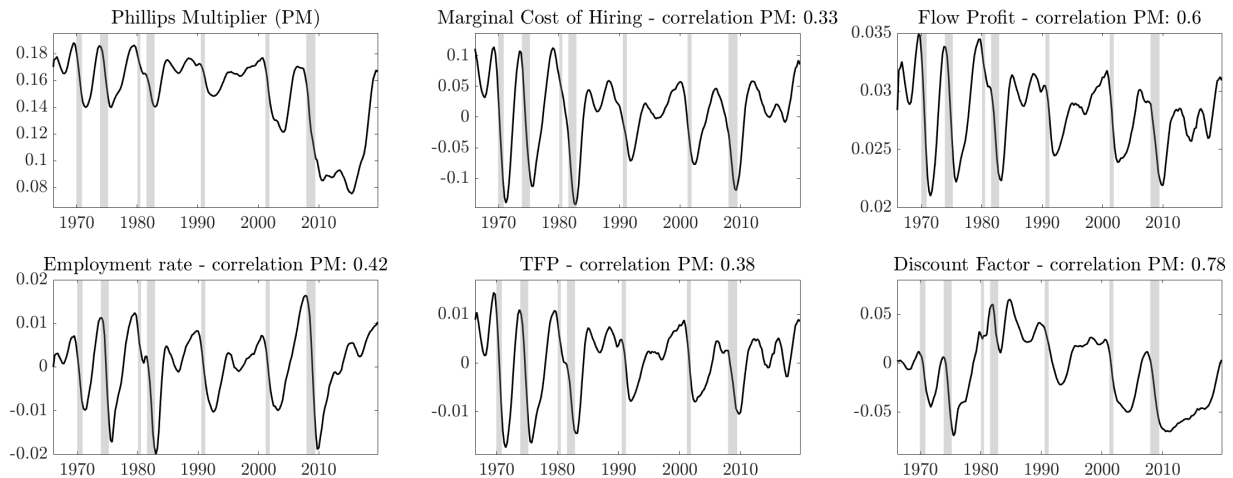
Using the model to filter the latent states, we compute the implied impulse response of a monetary policy shock at each point in time. These impulse responses also allow us to compute a time series for the Phillips Multiplier. We visualize this filtered series in Figure 8 until 2010, because this overlaps with our local projection sample, and for the full estimation sample in Figure 9. The average Phillips Multiplier estimate equals -0.15, consistent with Barnichon and Mesters (2021)

post-1990's estimate. Figure 8 shows the Phillips Multiplier is generally countercyclical, and, in absolute values, falls during recessions. Several periods stand out where the Phillips curve was particularly flat, typically following a recession, such as after the large recessions in the 70s, and after the great financial crisis. As can be seen in Figure 9, we observe that the Phillips Multiplier became particularly flat during the ZLB period, but after that has seen a steepening trend following the departure from the ZLB in the latter part of the sample.

We compare the model-implied Phillips Multiplier in Figure 8a with state-dependent local projection estimates from Figures 8b and 8c. Using unemployment as a proxy (Figure 8b), the empirical estimates closely track the model's implied cyclicity: the Phillips Multiplier falls during recessions and rises in expansions. While flow profit is not directly observed in the data, we use the model's filtered states to construct a corporate profit proxy and compare it to the corporate profit series from the data. As shown in Figure 8d, the model-implied and data-based proxies are positively correlated, despite corporate profit being an untargeted moment, providing an external validity check. Note that the data series leads the model-implied corporate profit by a few quarters, which helps explain why the corporate profit-based projections in Figure 8c show a Phillips Multiplier that leads both the model and the unemployment-based projections.

The fluctuations in the Phillips Multiplier can be interpreted through the mechanism outlined in the previous section. Figure 9 illustrates this by plotting the Phillips Multiplier alongside key proxies and state variables, with corresponding correlations reported. For instance, the model-implied marginal cost of hiring is moderately correlated with the Phillips Multiplier (0.33), as is total factor productivity (TFP) at 0.38. In contrast, flow profit shows a stronger correlation of 0.60, underscoring its central role in driving state dependence. Additionally, the Phillips Multiplier is influenced by variation in the discount factor, particularly in the latter part of the sample during the zero lower bound (ZLB) period.

Figure 9: Phillips Multiplier alongside Surplus Proxies and Model States



6 Extensions

This section considers three extensions that can be accommodated in our framework with minor adaptations.

6.1 Stochastic Volatility

A common extension of the standard framework introduces time-varying volatility in exogenous shocks, often modeled as stochastic volatility (SV). For instance, total factor productivity z_t may follow an AR(1)-SV process:

$$\begin{aligned} z_t &= \rho_z z_{t-1} + e^{\xi_t} u_t, \\ \xi_t &= (1 - \rho_\xi) \mu_\xi + \rho_\xi \xi_{t-1} + v_t, \end{aligned}$$

where ξ_t governs the volatility of u_t and follows an AR(1) with Gaussian innovations. The state vector now includes ξ_t . In Appendix F, we show how the Taylor Projection method accommodates SV, using the stochastic growth model as a tractable example. Attractively, the resulting solution is not certainty equivalent, with time-varying risk affecting policies. Achieving this with perturbation methods requires higher-order expansions (e.g., fourth or sixth order; see De Groot (2016)). Note that this provides an additional source of identification for the stochastic volatility; observables load on the volatility term directly.

Stochastic volatility complicates filtering due to nonlinear transition dynamics:

$$x_t = T x_t + c + R(x_t) \eta_t,$$

where $R(x_t)$ now depends on the state, introducing a nonlinearity that the Kalman filter cannot handle. In Appendix F, we discuss and illustrate two approaches: (i) the Extended Kalman Filter (EKF), linearizing $R(x_t)$ around $x_{t|t-1}$, or (ii) discretizing the process for ξ_t using, for example, the Rouwenhorst (1995) or Tauchen (1986) method, yielding a discrete process for $\tilde{\xi}_t$. Conditional on $\tilde{\xi}_t$, the system is linear and can be handled with the Kalman filter, while $\tilde{\xi}_t$ itself is filtered using the Hamilton filter (Hamilton, 1994). This approach builds on Friedman and Harris (1998) and Kitagawa (1996), with asymptotic justification provided by Farmer (2021).

6.2 Regime Switching or Rare Disasters

Another common extension is the regime-switching model, which underlies frameworks such as the rare disaster model (Barro, 2009), and also is used in Z. Liu et al. (2011), Foerster et al. (2016) and others. Here, the exogenous process departs from a stationary VAR(1), with dynamics instead governed by a latent discrete state $s_t \in 0, \dots, S$ evolving via a Markov chain with transition matrix

Π :

$$x_{2,t} = A(s_t)x_{2,t-1} + B(s_t)\eta_t, \quad \text{where } s_t \sim \Pi.$$

In Appendix F, we show how the Taylor Projection method can accommodate regime switching by assigning local regime-specific linear policy functions, $y_t = g_{s,x}(x_t)$, and transition functions, $x_{t+1} = h_{s,x}(x_t)$, for $s \in \{0, \dots, S\}$. Expectations of agents reflect that regime changes imply shifts in policy rules.

The model solution has a linear state-space representation conditional on a regime s_t , enabling likelihood evaluation via the Kalman filter. The latent state s_t is filtered using the Hamilton filter (Hamilton, 1994). As detailed in Appendix F, this approach is both fast and accurate. Unlike in the stochastic volatility case, the filtering here is not approximate but optimal.

6.3 Heterogeneous agents

Recall the notation for a standard Dynamic Stochastic General Equilibrium (DSGE) model in Equation (1):

$$\mathbb{E}_t [f(y_{t+1}, y_t, x_{t+1}, x_t)] = 0. \quad (29)$$

Heterogeneous agent models can also be written in this form, but the key challenge is that the state vector x_t includes an infinite-dimensional object: a measure μ_t over all idiosyncratic states. For example, in the heterogeneous household model of Krusell and Smith (1998), μ_t is the joint distribution over individual assets k and employment states ϵ . A common approach to reduce the state space is to approximate μ_t using dimension-reduction techniques such as binning (Reiter, 2009), parametrized distributions (Algan, Allais, and Den Haan, 2008), or moment-based polynomial expansions (Winberry, 2018).

For illustration, we focus on a simple heterogeneous household model; a detailed description is provided in Appendix G. In brief, the model features households $i \in [0, 1]$ with heterogeneous asset holdings and employment status $\epsilon \in \{0, 1\}$, which evolves exogenously according to a transition matrix $\Pi(z)$, where z denotes aggregate TFP, following an AR(1). Households supply labor inelastically and receive income $w_t[(1 - \tau_t)\epsilon + b(1 - \epsilon)]$, where w_t is the wage, τ_t the labor tax rate, and b unemployment benefits. Households save in risk-free assets a_{t+1} earning a return r_t , subject to a borrowing constraint $a_{t+1} \geq 0$.

A representative firm produces output using capital and labor, paying factor prices r_t and w_t , respectively. The model is closed by market-clearing conditions equating aggregate capital demand and supply, and aggregate output to total consumption plus investment. Since prices depend on the cross-sectional distribution of assets and employment, μ_t enters the equilibrium conditions, requiring households to forecast this distribution. As a result, μ_t is a state variable. Following Winberry (2018), we approximate infinitely-dimensional μ_t with a finite-dimensional vector of

moments, $\mathbf{m}_t$, as:

$$\mu_{\epsilon,t}(a) \approx g_{\epsilon,t}^0 \exp \left\{ g_{\epsilon,t}^1 (a - m_{\epsilon,t}^1) + \sum_{i=2}^{n_g} g_{\epsilon,t}^i [(a - m_{\epsilon,t}^1)^i - m_{\epsilon,t}^i] \right\}, \quad \text{for } \epsilon \in \{0, 1\}, \quad (30)$$

with $\mathbf{g}_t$ coefficients.

Besides approximating the cross-sectional distribution μ_t , to map this model in standard DSGE model form, we need to approximate the decision rules of the household, which follow from the household Euler equation and the borrowing constraint. Following Winberry (2018), we approximate the expected marginal utility from saving using Chebychev polynomials:

$$\mathbb{E} [\beta(1 + r_{t+1})c_{t+1}(\epsilon', a'_t(\epsilon, a))^{-\sigma}] \approx \exp \left\{ \sum_{i=1}^{n_\psi} \theta_{\epsilon,i,t} T_i(\xi(a)) \right\}.$$

with n_ψ the number of Chebychev nodes, T_i the i th-order Chebychev polynomial, θ_t are the coefficients, and $\xi(a) = 2 \frac{a-\underline{a}}{\bar{a}-\underline{a}} - 1$. The equality holds exactly at a selected set of nodes $\{a_j\}_{j=1}^{n_\psi}$.

This results in a minimal state-space representation with state vector $x_t = (z_t, \mathbf{m}_t)$ and controls $y_t = (\boldsymbol{\theta}_t, \mathbf{g}_t, r_t, w_t)$. The equilibrium conditions take the form of Equation (1), with the full system detailed in Appendix G. Rather than perturbing this system (Reiter, 2009; Winberry, 2018) or solving it via perfect foresight (Auclert et al., 2021), we use Taylor Projections. In what follows, when states and policies are expressed in log-deviations from steady state, they are denoted by $\hat{\cdot}$, and $\bar{\cdot}$ denotes a steady state value. In the Taylor Projection approach, if, for example, we use two moments used to approximate the cross-sectional distribution $\mu_{\epsilon,t}(a)$, the local linear policy functions take the form:

$$\begin{aligned} \hat{r}_t &= b_{0,x}^r + B_x^r \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}' \\ \hat{w}_t &= b_{0,x}^w + B_x^w \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}' \\ g_t &= \bar{g} + b_{0,x}^g + B_x^g \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}' \\ \theta_t &= \bar{\theta} + b_{0,x}^\theta + B_x^\theta \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}', \end{aligned}$$

where $\mathbf{g}_t$ and $\boldsymbol{\theta}_t$ are vectors, so B_x^g and B_x^θ are matrices. Under dynamic Taylor Projection, solving the model locally at different points in the state space yields different policy coefficients B_x^r , B_x^w , etc., allowing the solution to capture state dependence with respect to the current aggregate state z , as well as idiosyncratic distribution μ_t . Such state dependence has been shown to be empirically relevant. For instance, in heterogeneous firm models, Ottonello and Winberry (2020) show that monetary policy transmission varies with the distribution of firms' distance to default. Dynamic Taylor Projections can accommodate this type of state-dependent shock transmission.

In addition to the local linear policy rules, the Taylor Projection solution gives us linearized laws of motions for our moments, i.e.,

$$m_{\epsilon,t+1}^i = \bar{m}_\epsilon^i + b_{0,x}^{m,\epsilon,i} + B_x^{m,\epsilon,i} \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}', \quad \text{for } \epsilon \in \{0, 1\}, i = 1, 2, \dots \quad (31)$$

as well as other objects of interest, such as a linearized mapping for aggregate output

$$\hat{Y}_t = b_{0,x}^y + B_x^y \begin{bmatrix} \hat{z}_t & m_{0,t}^1 & m_{1,t}^1 & m_{0,t}^2 & m_{1,t}^2 \end{bmatrix}'$$

This allows us to map the model into a linear state space form. The state transition equation follows from Equation (31) and $\hat{z}_t$ being an AR(1). For the measurement equation, assume, for example, that we observe deviations of aggregate output, deviations of aggregate interest rates, and the second moment of asset holdings for both employed and unemployed (the choice of observables is not unique). Then, $y_t^{\text{obs}} = \begin{bmatrix} Y_t, r_t, m_{0,t}^2, m_{1,t}^2 \end{bmatrix}'$, with measurement equation

$$\begin{bmatrix} Y_t \\ r_t \\ m_{0,t}^2 \\ m_{1,t}^2 \end{bmatrix} = \begin{bmatrix} b_{0,x}^y \\ b_{0,x}^r \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} B_x^y & & & & \\ & B_x^r & & & \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{z}_t \\ m_{0,t}^1 \\ m_{1,t}^1 \\ m_{0,t}^2 \\ m_{1,t}^2 \end{bmatrix} + \varepsilon_t^{\text{ME}}, \quad \varepsilon_t^{\text{ME}} \sim N(0, Q),$$

where subscript "ss" refers to the steady state value. It is also possible to incorporate micro-level data, such as repeated cross-sections, into the estimation procedure, following the approach in L. Liu and Plagborg-Møller (2023). This involves evaluating the cross-sectional data points using the density in Equation (30) evaluated at a large number of draws from the model-implied distribution of the smoothed states $\mathbf{m}_t$ and $\mathbf{g}_t$. The resulting micro log-likelihood can then be combined with the macro log-likelihood derived from the filtered states.

7 Conclusion

This paper develops a computationally efficient method for full-information estimation of non-linear Dynamic Stochastic General Equilibrium (DSGE) models. Unlike conventional approaches that separate model solution from likelihood evaluation, our framework integrates the solution directly into the filtering process. This integration eliminates the need for computationally intensive global solution methods and simulation-based nonlinear filters, enabling accurate and tractable estimation of models with rich nonlinear features, including state-dependent shock transmission.

We illustrate the method using the Diamond-Mortensen-Pissarides (DMP) model, which is well known for its pronounced nonlinearities, and a standard New Keynesian framework. In both cases, the proposed approach achieves substantial computational gains: estimation times are com-

parable to those of linear methods and several orders of magnitude faster than existing nonlinear alternatives. In our main application, we show that search frictions in the labor market generate meaningful state dependence in the effects of monetary policy. The estimated Phillips Multiplier varies significantly over the business cycle, falling to roughly half its expansionary value during recessions.

The framework accommodates a range of extensions, including stochastic volatility, regime switching, and heterogeneous agent models. More broadly, the method offers a general and tractable approach to empirical macroeconomic analysis in settings where nonlinearities are essential, such as models with asset pricing, labor market frictions, nominal rigidities, or financial constraints.

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Supplementary Appendix to “Efficient Estimation of Nonlinear DSGE Models”

Eva F. Janssens

Sean McCrary

May, 2025

A Smoothing Algorithm

Algorithm 2 Time-Varying Kalman Smoother with Taylor Projections

Initialize: Set $t = T$, $r_T = 0$ and $N_T = 0$.

while $t \geq 1$ **do**

Update Step:

$$r_{t-1} = Z'_{\hat{s}_{t|t-1}} F_t^{-1} v_t + L'_t r_t \quad (\text{A.1})$$

$$N_{t-1} = Z'_{\hat{s}_{t|t-1}} F_t^{-1} Z_{\hat{s}_{t|t-1}} + L'_t N_t L_t \quad (\text{A.2})$$

where $L_t = T_{\hat{s}_{t|t-1}} - K_t Z_{\hat{s}_{t|t-1}}$, and K_t , v_t and F_t were stored from the Filter output.

Smoothing step:

 Compute the smoothed state estimate:

$$\hat{s}_{t|T} = a_{t|t-1} + P_{t|t-1} r_{t-1} \quad (\text{A.3})$$

 Compute the smoothed covariance:

$$P_{t|T} = P_{t|t-1} - P_{t|t-1} N_{t-1} P_{t|t-1} \quad (\text{A.4})$$

Set back time: Set $t \leftarrow t - 1$ and repeat.

end while

B Data

Table B1: Data details: estimation and filtering

Variable	Details	Transformation	Source
Employment $\tilde{n}_t$, one minus unemployment rate	Percent, Quarterly, Seasonally adjusted	log and HP-Filter, $\lambda = 1600$	UNRATE (FRED)
Market tightness $\tilde{\theta}_t$	Vacancies over unemployment, Quarterly	log and HP-Filter, $\lambda = 1600$	Barnicon (2010), JTSJOL, UNRATE (FRED)
Nominal rate $\tilde{r}_t$	3-Month Treasury Bill Secondary Market Rate, Discount Basis	Quarterly*, log	DTB3 (FRED)
Inflation $\tilde{\pi}_t$	Gross domestic product (implicit price deflator)	Quarter-to-quarter P_t/P_{t-1} , log	GDPDEF (FRED)
Job-finding rate $\tilde{f}_t$	Implied job-finding rate Shimer (2005)	Quarterly, log and HP-Filter, $\lambda = 1600$	UEMPLT5, URATE (FRED)

Notes: (*) R3M converted to a quarterly interest rate $(100 * ((1 + R3M/100)^{(1/4)} - 1))$.

C Textbook New-Keynesian Model

C.1 Households

There is a representative infinitely lived household that maximizes

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t D_t U(C_t, N_t) \right]$$

subject to a sequence of budget constraints

$$P_t C_t + B_t = R_{t-1} B_{t-1} + P_t W_t N_t + T_t$$

where C_t is consumption of the final good, B_t is an uncontingent nominal bond, R_{t-1} is the nominal interest rate, W_t is the real wage, and T_t contains dividends from firms net of taxes and transfers from the government. We assume utility has the following form

$$U(C_t, N_t) = \begin{cases} \frac{C_t^{1-\tau}-1}{1-\tau} - \chi \frac{N_t^{1+\eta}}{1+\eta}, & \text{if } \tau \neq 1 \\ \log(C_t) - \chi \frac{N_t^{1+\eta}}{1+\eta}, & \text{if } \tau = 1 \end{cases}$$

where $d_t = \log(D_t)$ is a random shock to the utility of consumption and follows an AR(1)

$$d_{t+1} = \rho_d d_t + \varepsilon_t^d, \quad \varepsilon_t^d \sim N(0, \sigma_d^2)$$

The optimal consumptions/saving and labor supply decisions are given by

$$\begin{aligned} \frac{\chi N_t^\eta}{C_t^{-\tau}} &= W_t \\ 1 &= \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right] \end{aligned}$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$. The stochastic discount factor is defined as

$$Q_{t,t+1} = \beta \frac{D_{t+1}}{D_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\tau}$$

C.2 Firms

Perfectly competitive final good firm combine a continuum of intermediate goods $j \in [0, 1]$.

Intermediate good j produced by a monopolist with technology $Y_t(j) = Z_t N_t(j)$ where Z_t is an aggregate technology process common to all firms. These firms choose prices $P_t(j)$ and labor

inputs $N_t(j)$ to maximize the present discounted value of profit

$$\mathbb{E}_0 \left[Q_{0,t} \left\{ \frac{P_t(j)}{P_t} Z_t N_t(j) - W_t N_t(j) - \frac{\phi}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right)^2 Y_t \right\} \right]$$

subject to a sequence of constraints

$$Z_t N_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-1/\gamma} Y_t$$

Optimal choices of $N_t(j)$ and $P_t(j)$ lead to the following condition

$$\begin{aligned} 0 = & \frac{Z_t N_t(j)}{P_t} - \phi \left(\frac{P_t(j)}{P_{t-1}(j)} - \Pi \right) \frac{Y_t}{P_{t-1,j}} - \frac{1}{\gamma} \left(\frac{P_t(j)}{P_t} - \frac{W_t}{Z_t} \right) \left(\frac{P_t(j)}{P_t} \right)^{-1/\gamma-1} \frac{Y_t}{P_t} \\ & + \phi \mathbb{E}_t \left[Q_{t,t+1} \left(\frac{P_{t+1}(j)}{P_t(j)} - \Pi \right) \frac{P_{t+1}(j)}{P_t(j)^2} Y_{t+1} \right] \end{aligned}$$

In a symmetric equilibrium $P_t(j) = P_t$ and $Y_t = Z_t N_t(j)$ for all j so this simplifies to

$$\frac{1}{\gamma} \left(1 - \frac{W_t}{Z_t} \right) = 1 - \phi (\Pi_t - \Pi) \Pi_t + \phi \mathbb{E}_t \left[Q_{t,t+1} (\Pi_{t+1} - \Pi) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right]$$

C.3 Government

The government levies lump-sum taxes and runs a balanced budget. The monetary authority follows a Taylor type rule

$$\frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{Y_t}{Y} \right)^{\phi_y} e^{m_t}$$

where m_t is an exogenous monetary policy shock that follows an AR(1)

$$m_{t+1} = \rho_m m_t + \varepsilon_t^m, \quad \varepsilon_t^m \sim N(0, \sigma_m^2)$$

C.4 Steady State

We focus on a non-stochastic steady state with inflation π equal to the Central Bank target which implies $R = \Pi/\beta$, and the parameter χ normalizes steady state output Y (that is N) to be unity $\chi = 1 - \gamma$. Note this implies $W = \chi$ so γ is the steady state wage markdown.

C.5 Equilibrium

The equilibrium conditions for market clearing in the labor and goods markets are

$$Y_t = C_t = N_t Z_t$$

$$W_t = (1 - \gamma) N_t^\eta C_t^\tau = (1 - \gamma) N_t^{\eta+\tau} Z_t^\tau$$

A minimal state representation consists of states $\mathbb{S} = \{z_t, d_t, m_t\}$ and policies $\{\Pi(\mathbb{S}), N(\mathbb{S})\}$ defined through the equilibrium conditions (substituting market clearing conditions where appropriate)

$$\text{Euler/IS : } 1 = \beta \mathbb{E}_t \left[e^{(\rho_d-1)d_t + \epsilon_{t+1}^d} \left(\frac{N_{t+1}}{N_t} e^{(\rho_z-1)z_t + \epsilon_{t+1}^z} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right]$$

$$\text{Pricing/NK-Phillips : } \frac{1}{\gamma} \left(1 - \chi N_t^{\eta+\tau} e^{(\tau-1)z_t} \right) = 1 - \phi (\Pi_t - \Pi) \Pi_t$$

$$+ \phi \beta \mathbb{E}_t \left[e^{(\rho_d-1)d_t + \epsilon_{t+1}^d} \left(\frac{N_{t+1}}{N_t} e^{(\rho_z-1)z_t + \epsilon_{t+1}^z} \right)^{1-\tau} (\Pi_{t+1} - \Pi) \Pi_{t+1} \right]$$

and an auxiliary equation

$$\text{Taylor Rule : } \frac{R_t}{R} = \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{Y} \right)^{\phi_y} e^{m_t}$$

Using the steady state relations and letting lower case letters denote log-deviations from steady state values $x = \ln X_t - \ln X$, the equilibrium conditions become

$$1 = \mathbb{E}_t \left[e^{(\rho_d-1)d_t + \epsilon_{t+1}^d - \tau(n_{t+1} - n_t + (\rho_z-1)z_t + \epsilon_{t+1}^z) + r_t - \pi_{t+1}} \right]$$

$$\kappa \left(1 - e^{(\eta+\tau)n_t + (\tau-1)z} \right) = - (e^{\pi_t} - 1) e^{\pi_t} + \beta \mathbb{E}_t \left[e^{(\rho_d-1)d_t + \epsilon_{t+1}^d + (1-\tau)(n_{t+1} - n_t + (\rho_z-1)z_t + \epsilon_{t+1}^z)} (e^{\pi_{t+1}} - 1) e^{\pi_{t+1}} \right]$$

$$r_t = \phi_\pi \pi_t + \phi_y (n_t + z + t) + m_t$$

where $\kappa = \frac{1-\gamma}{\phi \Pi^2 \gamma}$

C.6 Log-linear System

Lowercase signifies log-deviation from steady state

$$\begin{aligned}
n_t &= \mathbb{E}_t[n_{t+1}] + \frac{1}{\tau}(1 - \rho_d)d_t - (1 - \rho_z)z_t - \frac{1}{\tau}\{r_t - \mathbb{E}_t[\pi_{t+1}]\} \\
\pi_t &= \beta\mathbb{E}_t[\pi_{t+1}] + \kappa[(\eta + \tau)n_t + (\tau - 1)z_t] \\
r_t &= \phi_\pi\pi_t + \phi_y(n_t + z_t) + m_t \\
m_t &= \rho_m m_{t-1} + \varepsilon_t^m \\
z_t &= \rho_z z_{t-1} + \varepsilon_t^z \\
d_t &= \rho_d d_{t-1} + \varepsilon_t^d
\end{aligned}$$

C.7 Local Approximation

Note, we can write the variables of the equilibrium system in terms of log deviations from steady state (lowercase signifies logs, e.g, $\pi_t = \ln \Pi_t - \ln \Pi$), after substituting the Taylor Rule into the Euler equation, as

$$\begin{aligned}
1 &= \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \varepsilon_{t+1}^d - \tau\{n_{t+1} - n_t + (\rho_z - 1)z_t + \varepsilon_{t+1}^z\} + \phi_\pi\pi_t + \phi_y(n_t + z_t) + m_t - \pi_{t+1}} \right] \\
\frac{1 - \gamma}{\Pi^2 \phi \gamma} \left(1 - e^{(\eta + \tau)n_t + (\tau - 1)z_t} \right) &= - (e^{\pi_t} - 1) e^{\pi_t} \\
&\quad + \beta \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \varepsilon_{t+1}^d + (1 - \tau)\{n_{t+1} - n_t + (\rho_z - 1)z_t + \varepsilon_{t+1}^z\}} (e^{\pi_{t+1}} - 1) e^{\pi_{t+1}} \right]
\end{aligned}$$

Which can be simplified to

$$\begin{aligned}
1 &= e^{(\rho_d - 1)d_t + \tau n_t + \tau(1 - \rho_z)z_t + \phi_\pi\pi_t + \phi_y(n_t + z_t) + m_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau n_{t+1} - \tau \varepsilon_{t+1}^z - \pi_{t+1}} \right] \\
\frac{1 - \gamma}{\Pi^2 \phi \gamma} \left(1 - e^{(\eta + \tau)n_t + (\tau - 1)z_t} \right) &= - e^{2\pi_t} + e^{\pi_t} \\
&\quad + \beta \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \varepsilon_{t+1}^d + (1 - \tau)\{n_{t+1} - n_t + (\rho_z - 1)z_t + \varepsilon_{t+1}^z\} + 2\pi_{t+1}} \right] \\
&\quad - \beta \mathbb{E}_t \left[e^{(\rho_d - 1)d_t + \varepsilon_{t+1}^d + (1 - \tau)\{n_{t+1} - n_t + (\rho_z - 1)z_t + \varepsilon_{t+1}^z\} + \pi_{t+1}} \right]
\end{aligned}$$

The first step is to suppose the policies are linear in logs

$$\begin{aligned}
\pi_t &= a_0 + a_1 z_t + a_2 d_t + a_3 m_t \\
n_t &= b_0 + b_1 z_t + b_2 d_t + b_3 m_t
\end{aligned}$$

This implies the future policies are of the form

$$\begin{aligned}\pi_{t+1} &= a_0 + a_1 \rho_z z_t + a_1 \epsilon_{t+1}^z + a_2 \rho_d d_t + a_2 \epsilon_{t+1}^d + a_3 \rho_m m_t + a_3 \epsilon_{t+1}^m \\ n_{t+1} &= b_0 + b_1 \rho_z z_t + b_1 \epsilon_{t+1}^z + b_2 \rho_d d_t + b_2 \epsilon_{t+1}^d + b_3 \rho_m m_t + b_3 \epsilon_{t+1}^m\end{aligned}$$

Moreover, it implies

$$n_{t+1} - n_t = b_1(\rho_z - 1)z_t + b_1\epsilon_{t+1}^z + b_2(\rho_d - 1)d_t + b_2\epsilon_{t+1}^d + b_3(\rho_m - 1)m_t + b_3\epsilon_{t+1}^m$$

The goal of the local approximation method is to solve for the policy coefficients $\mathbf{a}$ and $\mathbf{b}$ as implicit functions of the current state (z_t, d_t, m_t) .

Euler equation. Consider the Euler equation with the policies substituted

$$\begin{aligned}1 &= e^{(\rho_d - 1)d_t - \tau[b_1(\rho_z - 1)z_t + b_2(\rho_d - 1)d_t + b_3(\rho_m - 1)m_t] - \tau(\rho_z - 1)z_t + r_t - a_1 - a_1\rho_z z_t - a_2\rho_d d_t - a_3\rho_m m_t} \\ &\cdot \mathbb{E}_t \left[e^{\epsilon_{t+1}^d - \tau(b_1\epsilon_{t+1}^z + b_2\epsilon_{t+1}^d + b_3\epsilon_{t+1}^m) - \tau\epsilon_{t+1}^z - a_1\epsilon_{t+1}^z - a_2\epsilon_{t+1}^d - a_3\epsilon_{t+1}^m} \right]\end{aligned}$$

Collecting terms in the expectation yields

$$\mathbb{E}_t \left[e^{\epsilon_{t+1}^d - \tau(b_1\epsilon_{t+1}^z + b_2\epsilon_{t+1}^d + b_3\epsilon_{t+1}^m) - \tau\epsilon_{t+1}^z - a_1\epsilon_{t+1}^z - a_2\epsilon_{t+1}^d - a_3\epsilon_{t+1}^m} \right] = \mathbb{E}_t \left[e^{(1 - \tau b_2 - a_2)\epsilon_{t+1}^d - (\tau(b_1 + 1) + a_1)\epsilon_{t+1}^z - \tau(b_3 + a_3)\epsilon_{t+1}^m} \right]$$

For a mean zero normal random variable X , and constant c , the expectation satisfies

$$\mathbb{E} \left[e^{cX} \right] = e^{\frac{1}{2}c^2\sigma_x^2},$$

so the Euler equation simplifies to

$$\begin{aligned}1 &= e^{(\rho_d - 1)d_t - \tau[b_1(\rho_z - 1)z_t + b_2(\rho_d - 1)d_t + b_3(\rho_m - 1)m_t] - \tau(\rho_z - 1)z_t + r_t - a_1 - a_1\rho_z z_t - a_2\rho_d d_t - a_3\rho_m m_t} \\ &\cdot e^{\frac{1}{2}\{(1 - \tau b_2 - a_2)^2\sigma_d^2 + (\tau(b_1 + 1) + a_1)^2\sigma_z^2 + \tau^2(b_3 + a_3)^2\sigma_m^2\}}\end{aligned}$$

Note, by taking logs and substituting the Taylor Rule, this is equivalent to

$$\begin{aligned}0 &= (\rho_d - 1)d_t + \tau[b_1(1 - \rho_z)z_t + b_2(1 - \rho_d)d_t + b_3(1 - \rho_m)m_t] + \tau(1 - \rho_z)z_t \\ &+ \phi_\pi(a_0 + a_1 z_t + a_2 d_t + a_3 m_t) + \phi_y(b_0 + b_1 z_t + b_2 d_t + b_3 m_t + z_t) + m_t \\ &- (a_0 + a_1 \rho_z z_t + a_2 \rho_d d_t + a_3 \rho_m m_t) \\ &+ \frac{1}{2}\{(1 - \tau b_2 - a_2)^2\sigma_d^2 + (\tau(b_1 + 1) + a_1)^2\sigma_z^2 + \tau^2(b_3 + a_3)^2\sigma_m^2\}\end{aligned}$$

which implies the variance terms show up in the Euler equation, but not the partial derivatives of the Euler equation with respect to the states. The partial derivatives w.r.t. the policy coefficients are given by

$$\begin{aligned}
\partial b_0 &: \phi_y \\
\partial b_1 &: [\tau(1 - \rho_z) + \phi_y] z_t + \tau(\tau(b_1 + 1) + a_1) \sigma_z^2 \\
\partial b_2 &: [\tau(1 - \rho_d) + \phi_y] d_t - \tau(1 - \tau b_2 - a_2) \sigma_d^2 \\
\partial b_3 &: [\tau(1 - \rho_m) + \phi_y] m_t + \tau^2(b_3 + a_3) \sigma_m^2 \\
\partial a_0 &: \phi_p - 1 \\
\partial a_1 &: (\phi_p - \rho_z) z_t + (\tau(b_1 + 1) + a_1) \sigma_z^2 \\
\partial a_2 &: (\phi_p - \rho_d) d_t - (1 - \tau b_2 - a_2) \sigma_d^2 \\
\partial a_3 &: (\phi_p - \rho_m) m_t + \tau^2(b_3 + a_3) \sigma_m^2
\end{aligned}$$

Moreover, the partial derivatives with respect to the states are given by

$$\begin{aligned}
\partial z_t : 0 &= \tau b_1 + \tau(1 - \rho_z) + \phi_\pi a_1 + \phi_y b_1 - \tau b_1 \rho_z - a_1 \rho_z \\
\Rightarrow 0 &= \tau(1 - \rho_z) + [\tau(1 - \rho_z) + \phi_y] b_1 + (\phi_\pi - \rho_z) a_1 \\
\partial z_t \partial b_1 &: \tau(1 - \rho_z) + \phi_y \\
\partial z_t \partial a_1 &: \phi_\pi - \rho_z
\end{aligned}$$

$$\begin{aligned}
\partial d_t : 0 &= (\rho_d - 1) + \tau b_2 + \phi_\pi a_2 + \phi_y b_2 - \tau b_2 \rho_d - a_2 \rho_d \\
\Rightarrow 0 &= (\rho_d - 1) + [\tau(1 - \rho_d) + \phi_y] b_2 + (\phi_\pi - \rho_d) a_2 \\
\partial d_t \partial b_2 &: \tau(1 - \rho_d) + \phi_y \\
\partial d_t \partial a_2 &: \phi_\pi - \rho_d
\end{aligned}$$

$$\begin{aligned}
\partial m_t : 0 &= \tau b_3 + \phi_\pi a_3 + \phi_y b_3 + 1 - \tau b_3 \rho_m - a_3 \rho_m \\
\Rightarrow 0 &= 1 + [\tau(1 - \rho_m) + \phi_y] b_3 + (\phi_\pi - \rho_m) a_3 \\
\partial m_t \partial b_3 &: \tau(1 - \rho_m) + \phi_y \\
\partial m_t \partial a_3 &: \phi_\pi - \rho_m
\end{aligned}$$

which gives the residual function derivatives and elements of the Jacobian related to the Euler equation. Note, only the elements of the Jacobian that depend on the policy coefficients a and b need to be updated in the Newton solution step. Of the 64 elements of the Jacobian, 32 are

determined by the Euler equation block, of which *only* 6 depend on the policy coefficients.

Zero Lower Bound. We are interested in models where there may be a binding zero lower bound on the nominal interest rate. Formally,

$$\begin{aligned} R_t &= \max \{1, R_t^s\} \\ \Rightarrow \frac{R_t}{R} &= \max \left\{ \frac{1}{R}, \frac{R_t^s}{R} \right\} \\ \frac{R_t^s}{R} &= \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{Y} \right)^{\phi_y} e^{m_t} \end{aligned}$$

where R_t^s is the “shadow-rate” determined by a Taylor type rule. Taking log deviation from steady-state values $r_t = \ln R_t - \ln R$ and $r_t^s = \ln R_t^s - \ln R$, and denoting $r = \ln R$, we have

$$\begin{aligned} r_t &= \max \{-r, r_t^s\} \\ r_t^s &= \phi_\pi \pi_t + \phi_y (n_t + z_t) + m_t \end{aligned}$$

To deal with the ZLB we proceed by using a smooth approximation to the max operator. Specifically, we use the following

$$r_t = \max \{-r, r_t^s\} \approx \frac{1}{\lambda} \ln \left(e^{-\lambda r} + e^{\lambda r_t^s} \right),$$

where λ is a parameter. Note, as $\lambda \rightarrow \infty$, the approximation becomes exact. Moreover, we have

$$e^{r_t} = \left(e^{-\lambda r} + e^{\lambda r_t^s} \right)^{1/\lambda}$$

Using the expression from before for the Euler equation we have

$$\begin{aligned} 1 &= \left(e^{-\lambda r} + e^{\lambda (\phi_\pi (a_0 + a_1 z_t + a_2 d_t + a_3 m_t) + \phi_y (b_0 + b_1 z_t + b_2 d_t + b_3 m_t + z_t) + m_t)} \right)^{1/\lambda} \\ &\quad e^{(\rho_d - 1)d_t - \tau [b_1(\rho_z - 1)z_t + b_2(\rho_d - 1)d_t + b_3(\rho_m - 1)m_t] - \tau(\rho_z - 1)z_t - a_1 - a_1 \rho_z z_t - a_2 \rho_d d_t - a_3 \rho_m m_t} \\ &\quad \cdot e^{\frac{1}{2} \{ (1 - \tau b_2 - a_2)^2 \sigma_d^2 + (\tau(b_1 + 1) + a_1)^2 \sigma_z^2 + \tau^2 (b_3 + a_3)^2 \sigma_m^2 \}}, \end{aligned}$$

so we can again eliminate the expectation. The pricing equation is not directly affected by interest rates, so that block is unchanged.

Pricing equation. Now consider the pricing equation with the policies substituted

$$\begin{aligned}
& \frac{1-\gamma}{\Pi^2\phi\gamma} \left(1 - e^{(\eta+\tau)(b_0+b_1z_t+b_2d_t+b_3m_t)+(\tau-1)z_t} \right) = -e^{2(a_0+a_1z_t+a_2d_t+a_3m_t)} + e^{a_0+a_1z_t+a_2d_t+a_3m_t} \\
& + \beta e^{(\rho_d-1)d_t+(1-\tau)\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\}+2(a_0+a_1\rho_zz_t+a_2\rho_dd_t+a_3\rho_mm_t)} \\
& \cdot \mathbb{E}_t \left[e^{\epsilon_{t+1}^d+(1-\tau)\{b_1\epsilon_{t+1}^z+b_2\epsilon_{t+1}^d+b_3\epsilon_{t+1}^m+\epsilon_{t+1}^z\}+2(a_1\epsilon_{t+1}^z+a_2\epsilon_{t+1}^d+a_3\epsilon_{t+1}^m)} \right] \\
& - \beta e^{(\rho_d-1)d_t+(1-\tau)\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\}+a_0+a_1\rho_zz_t+a_2\rho_dd_t+a_3\rho_mm_t} \\
& \cdot \mathbb{E}_t \left[e^{\epsilon_{t+1}^d+(1-\tau)\{b_1\epsilon_{t+1}^z+b_2\epsilon_{t+1}^d+b_3\epsilon_{t+1}^m+\epsilon_{t+1}^z\}+a_1\epsilon_{t+1}^z+a_2\epsilon_{t+1}^d+a_3\epsilon_{t+1}^m} \right]
\end{aligned}$$

focusing on the expectation terms, we have the following

$$\begin{aligned}
& \mathbb{E}_t \left[e^{\epsilon_{t+1}^d+(1-\tau)\{b_1\epsilon_{t+1}^z+b_2\epsilon_{t+1}^d+b_3\epsilon_{t+1}^m+\epsilon_{t+1}^z\}+2(a_1\epsilon_{t+1}^z+a_2\epsilon_{t+1}^d+a_3\epsilon_{t+1}^m)} \right] \\
& = \mathbb{E}_t \left[e^{(1+(1-\tau)b_2+2a_2)\epsilon_{t+1}^d+((1-\tau)(b_1+1)+2a_1)\epsilon_{t+1}^z+((1-\tau)b_3+2a_3)\epsilon_{t+1}^m} \right] \\
& = e^{\frac{1}{2}\{(1+(1-\tau)b_2+2a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+2a_1)^2\sigma_z^2+((1-\tau)b_3+2a_3)^2\sigma_m^2\}} \\
& \mathbb{E}_t \left[e^{\epsilon_{t+1}^d+(1-\tau)\{b_1\epsilon_{t+1}^z+b_2\epsilon_{t+1}^d+b_3\epsilon_{t+1}^m+\epsilon_{t+1}^z\}+a_1\epsilon_{t+1}^z+a_2\epsilon_{t+1}^d+a_3\epsilon_{t+1}^m} \right] \\
& = \mathbb{E}_t \left[e^{(1+(1-\tau)b_2+a_2)\epsilon_{t+1}^d+((1-\tau)(b_1+1)+a_1)\epsilon_{t+1}^z+((1-\tau)b_3+a_3)\epsilon_{t+1}^m} \right] \\
& = e^{\frac{1}{2}\{(1+(1-\tau)b_2+a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+a_1)^2\sigma_z^2+((1-\tau)b_3+a_3)^2\sigma_m^2\}}
\end{aligned}$$

which implies the pricing equation simplifies to

$$\begin{aligned}
& \kappa \left(1 - e^{(\eta+\tau)(b_0+b_1z_t+b_2d_t+b_3m_t)+(\tau-1)z_t} \right) = -e^{2(a_0+a_1z_t+a_2d_t+a_3m_t)} + e^{a_0+a_1z_t+a_2d_t+a_3m_t} \\
& + \beta e^{(\rho_d-1)d_t+(1-\tau)\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\}+2(a_0+a_1\rho_zz_t+a_2\rho_dd_t+a_3\rho_mm_t)} \\
& \cdot e^{\frac{1}{2}\{(1+(1-\tau)b_2+2a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+2a_1)^2\sigma_z^2+((1-\tau)b_3+2a_3)^2\sigma_m^2\}} \\
& - \beta e^{(\rho_d-1)d_t+(1-\tau)\{b_1(\rho_z-1)z_t+b_2(\rho_d-1)d_t+b_3(\rho_m-1)m_t+(\rho_z-1)z_t\}+a_0+a_1\rho_zz_t+a_2\rho_dd_t+a_3\rho_mm_t} \\
& \cdot e^{\frac{1}{2}\{(1+(1-\tau)b_2+a_2)^2\sigma_d^2+((1-\tau)(b_1+1)+a_1)^2\sigma_z^2+((1-\tau)b_3+a_3)^2\sigma_m^2\}}
\end{aligned}$$

where $\kappa = \frac{1-\gamma}{\Pi^2\phi\gamma}$. Here, the derivatives of the residual function and elements of the Jacobian are more elaborate, but given that this is an analytic expression we can rely on symbolic differentiation to obtain the results.

D NK-DMP Model

Here we describe the NK-DMP model. The equilibrium conditions are

$$\begin{aligned} 1 &= \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{-\tau} \frac{R_t}{\Pi_{t+1}} \right] \\ \frac{\Omega_t}{\gamma} &= 1 - \phi (\Pi_t - \Pi) \Pi_t + \phi \beta \mathbb{E}_t \left[\frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{1-\tau} (\Pi_{t+1} - \Pi) \Pi_{t+1} \right] \\ \frac{c}{Q_t} &= (1 - \eta) ((1 - \Omega_t) e^{z_t} - \nu) + \beta \mathbb{E}_t \left[(1 - \delta_{t+1}) \frac{D_{t+1}}{D_t} \left(\frac{N_{t+1} Z_{t+1}}{N_t Z_t} \right)^{-\tau} \frac{c}{Q_{t+1}} \right] \end{aligned}$$

with auxiliary equations

$$\begin{aligned} \frac{R_t}{R} &= \left(\frac{\Pi_t}{\Pi} \right)^{\phi_\pi} \left(\frac{N_t Z_t}{N} \right)^{\phi_y} e^{m_t} \\ N_t &= (1 - \delta_t) N_{t-1} + Q_t \Theta_t (1 - (1 - \delta_t) N_{t-1}) \\ Q_t &= A_t \Theta_t^{-\alpha} \\ \delta_t &= \delta e^{s_t} \\ A_t &= A e^{a_t} \\ z_t &= \rho_z z_{t-1} + \epsilon_t^z \\ d_t &= \rho_d d_{t-1} + \epsilon_t^d \\ m_t &= \rho_m m_{t-1} + \epsilon_t^m \\ s_t &= \rho_s s_{t-1} + \epsilon_t^s \\ a_t &= \rho_a a_{t-1} + \epsilon_t^a \end{aligned}$$

where $z_t = \ln Z_t$ and $d_t = \ln D_t$. A minimal state representation has six states $\mathbb{S} = \{N_{t-1}, z_t, d_t, m_t, s_t, a_t\}$ and three policies $\{\theta(\mathbb{S}), \Pi(\mathbb{S}), \omega(\mathbb{S})\}$

D.1 Steady State

We focus on a non-stochastic steady state where inflation is at the central bank target Π . This implies $\beta = \Pi/R$ and $\omega = \gamma$. In the labor market, we normalize $\theta = 1$, and $q = 0.7$, and $N = 0.945$. This implies $A = 0.7$ and

$$\delta N = q\theta - q\theta(1 - \delta)N \Rightarrow \delta = \frac{1 - N}{N} \frac{q\theta}{1 - q\theta} \approx 0.1358$$

D.2 Local Approximation

In what follows it will be useful to write variables X_t in terms of log deviations from their steady state values $x_t = \ln X_t - \ln X$. Moreover we will add a linear policy for employment n_t .

This yields an equilibrium system

$$\begin{aligned}
1 &= \mathbb{E}_t \left[e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)+r_t-\pi_{t+1}} \right] \\
\frac{1}{\phi \Pi^2} (e^{\omega_t} - 1) &= -(e^{\pi_t} - 1) e^{\pi_t} + \beta \mathbb{E}_t \left[e^{d_{t+1}-d_t+(1-\tau)(n_{t+1}+z_{t+1}-n_t-z_t)} (e^{\pi_{t+1}} - 1) e^{\pi_{t+1}} \right] \\
e^{-q_t} &= \frac{Q(1-\eta)}{c} ((1-\gamma e^{\omega_t}) e^{z_t} - v) + \beta \mathbb{E}_t \left[(1-\delta e^{s_{t+1}}) e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}} \right] \\
e^{n_t-n_{t-1}} &= (1-\delta e^{s_t}) + Q \theta e^{q_t+\theta_t} \left(\frac{1}{N} e^{-n_{t-1}} - (1-\delta e^{s_t}) \right)
\end{aligned}$$

where

$$\begin{aligned}
\theta_t &= b_{10} + b_{11}n_{t-1} + b_{12}z_t + b_{13}d_t + b_{14}m_t + b_{15}s_t + b_{16}a_t \\
\pi_t &= b_{20} + b_{21}n_{t-1} + b_{22}z_t + b_{23}d_t + b_{24}m_t + b_{25}s_t + b_{26}a_t \\
\omega_t &= b_{30} + b_{31}n_{t-1} + b_{32}z_t + b_{33}d_t + b_{34}m_t + b_{35}s_t + b_{36}a_t \\
n_t &= b_{40} + b_{41}n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t \\
q_t &= a_t - \alpha \theta_t \\
r_t &= \phi_\pi \pi_t + \phi_y (n_t + z_t) + m_t
\end{aligned}$$

Which implies

$$n_t - n_{t-1} = b_{40} + (b_{41} - 1)n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t$$

$$n_{t+1} - n_t = b_{41}(n_t - n_{t-1}) + b_{42}(z_{t+1} - z_t) + b_{43}(d_{t+1} - d_t) + b_{44}(m_{t+1} - m_t) + b_{45}(s_{t+1} - s_t) + b_{46}(a_{t+1} - a_t)$$

$$\begin{aligned}
&= b_{41}(n_t - n_{t-1}) + b_{42}(\rho_z - 1)z_t + b_{43}(\rho_d - 1)d_t + b_{44}(\rho_m - 1)m_t + b_{45}(\rho_s - 1)s_t + b_{46}(\rho_a - 1)a_t \\
&\quad + b_{42}\varepsilon_{t+1}^z + b_{43}\varepsilon_{t+1}^d + b_{44}\varepsilon_{t+1}^m + b_{45}\varepsilon_{t+1}^s + b_{46}\varepsilon_{t+1}^a
\end{aligned}$$

$$\begin{aligned}
&= b_{41}(b_{40} + (b_{41} - 1)n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t) \\
&\quad + b_{42}(\rho_z - 1)z_t + b_{43}(\rho_d - 1)d_t + b_{44}(\rho_m - 1)m_t + b_{45}(\rho_s - 1)s_t + b_{46}(\rho_a - 1)a_t \\
&\quad + b_{42}\varepsilon_{t+1}^z + b_{43}\varepsilon_{t+1}^d + b_{44}\varepsilon_{t+1}^m + b_{45}\varepsilon_{t+1}^s + b_{46}\varepsilon_{t+1}^a
\end{aligned}$$

$$\begin{aligned}
&= b_{41}b_{40} + b_{41}(b_{41} - 1)n_{t-1} \\
&\quad + b_{42}(\rho_z - 1 + b_{41})z_t + b_{43}(\rho_d - 1 + b_{41})d_t + b_{44}(\rho_m - 1 + b_{41})m_t + b_{45}(\rho_s - 1 + b_{41})s_t + b_{46}(\rho_a - 1 + b_{41})a_t \\
&\quad + b_{42}\varepsilon_{t+1}^z + b_{43}\varepsilon_{t+1}^d + b_{44}\varepsilon_{t+1}^m + b_{45}\varepsilon_{t+1}^s + b_{46}\varepsilon_{t+1}^a
\end{aligned}$$

Note, $n_{t+1} - n_t$ can be separated into two parts (i) terms determined at time t , and (ii) shocks that arrive at $t + 1$.

$$n_{t+1} - n_t = B_0^n + B_1^n x_t + B_2^n \epsilon_{t+1}$$

Similarly, π_{t+1} , and the exogenous shocks can be written as

$$\begin{aligned} \pi_{t+1} &= b_{20} + b_{21}n_t + b_{22}z_{t+1} + b_{23}d_{t+1} + b_{24}m_{t+1} + b_{25}s_{t+1} + b_{26}a_{t+1} \\ &= b_{20} + b_{21}(b_{40} + b_{41}n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t) \\ &\quad + b_{22}\rho_z z_t + b_{23}\rho_d d_t + b_{24}\rho_m m_t + b_{25}\rho_s s_t + b_{26}\rho_a a_t \\ &\quad + b_{22}\epsilon_{t+1}^z + b_{23}\epsilon_{t+1}^d + b_{24}\epsilon_{t+1}^m + b_{25}\epsilon_{t+1}^s + b_{26}\epsilon_{t+1}^a \\ &= B_0^\pi + B_1^\pi x_t + B_2^\pi \epsilon_{t+1} \end{aligned}$$

$$z_{t+1} - z_t = (\rho_z - 1)z_t + \epsilon_{t+1}^z$$

$$d_{t+1} - d_t = (\rho_d - 1)d_t + \epsilon_{t+1}^d$$

Finally θ_{t+1} and q_{t+1} can be written as

$$\begin{aligned} \theta_{t+1} &= b_{10} + b_{11}n_t + b_{12}z_{t+1} + b_{13}d_{t+1} + b_{14}m_{t+1} + b_{15}s_{t+1} + b_{16}a_{t+1} \\ &= b_{10} + b_{11}(b_{40} + b_{41}n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t) \\ &\quad + b_{12}\rho_z z_t + b_{13}\rho_d d_t + b_{14}\rho_m m_t + b_{15}\rho_s s_t + b_{16}\rho_a a_t \\ &\quad + b_{12}\epsilon_{t+1}^z + b_{13}\epsilon_{t+1}^d + b_{14}\epsilon_{t+1}^m + b_{15}\epsilon_{t+1}^s + b_{16}\epsilon_{t+1}^a \\ &= B_0^\theta + B_1^\theta x_t + B_2^\theta \epsilon_{t+1} \\ q_{t+1} &= \rho_a a_t + \epsilon_{t+1}^a - \alpha B_0^\theta - \alpha B_1^\theta x_t - \alpha B_2^\theta \epsilon_{t+1} \end{aligned}$$

Euler Equation. Consider the Euler equation

$$1 = \mathbb{E}_t \left[e^{d_{t+1} - d_t - \tau(n_{t+1} + z_{t+1} - n_t - z_t) + r_t - \pi_{t+1}} \right]$$

Using the previous expressions, we can write the Euler equation compactly as

$$1 = e^{(\rho_d-1)d_t - \tau(\rho_z-1)z_t - \tau B_0^n - \tau B_1^n x_t + r_t - B_0^\pi - B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z - B_2^\pi \varepsilon_{t+1}} \right]$$

and the expectation can be expressed in closed form as

$$\begin{aligned} & \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z - B_2^\pi \varepsilon_{t+1}} \right] \\ &= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau(b_{42}\varepsilon_{t+1}^z + b_{43}\varepsilon_{t+1}^d + b_{44}\varepsilon_{t+1}^m + b_{45}\varepsilon_{t+1}^s + b_{46}\varepsilon_{t+1}^a) - \tau\varepsilon_{t+1}^z - (b_{22}\varepsilon_{t+1}^z + b_{23}\varepsilon_{t+1}^d + b_{24}\varepsilon_{t+1}^m + b_{25}\varepsilon_{t+1}^s + b_{26}\varepsilon_{t+1}^a)} \right] \\ &= \mathbb{E}_t \left[e^{(1-\tau b_{43}-b_{23})\varepsilon_{t+1}^d - (\tau(b_{42}+1)+b_{22})\varepsilon_{t+1}^z - (\tau b_{44}+b_{24})\varepsilon_{t+1}^m - (\tau b_{45}+b_{25})\varepsilon_{t+1}^s - (\tau b_{46}+b_{26})\varepsilon_{t+1}^a} \right] \\ &= e^{\frac{1}{2}((1-\tau b_{43}-b_{23})^2\sigma_d^2 + (\tau(b_{42}+1)+b_{22})^2\sigma_z^2 + (\tau b_{44}+b_{24})^2\sigma_m^2 + (\tau b_{45}+b_{25})^2\sigma_s^2 + (\tau b_{46}+b_{26})^2\sigma_a^2)} \end{aligned}$$

Pricing equation Consider the optimal price setting condition

$$\frac{1}{\phi\Pi^2} (e^{\omega_t} - 1) = -(e^{\pi_t} - 1) e^{\pi_t} + \beta \mathbb{E}_t \left[e^{d_{t+1}-d_t+(1-\tau)(n_{t+1}+z_{t+1}-n_t-z_t)} (e^{\pi_{t+1}} - 1) e^{\pi_{t+1}} \right]$$

the continuation value can be solved in closed form given the linear policies, first note

$$\begin{aligned} & \mathbb{E}_t \left[e^{d_{t+1}-d_t+(1-\tau)(n_{t+1}+z_{t+1}-n_t-z_t)} (e^{\pi_{t+1}} - 1) e^{\pi_{t+1}} \right] \\ &= \mathbb{E}_t \left[e^{d_{t+1}-d_t+(1-\tau)(n_{t+1}+z_{t+1}-n_t-z_t)} (e^{2\pi_{t+1}} - e^{\pi_{t+1}}) \right] \\ &= e^{(\rho_d-1)d_t+(1-\tau)[(\rho_z-1)z_t+B_0^n+B_1^n x_t]} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)} (e^{2\pi_{t+1}} - e^{\pi_{t+1}}) \right] \end{aligned}$$

and the remaining expectation term can be written as

$$\mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+2\pi_{t+1}} \right] - \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+\pi_{t+1}} \right]$$

where

$$\begin{aligned} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+2\pi_{t+1}} \right] &= e^{2B_0^\pi + 2B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+2B_2^\pi \varepsilon_{t+1}} \right] \\ \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+\pi_{t+1}} \right] &= e^{B_0^\pi + B_1^\pi x_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d+(1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z)+B_2^\pi \varepsilon_{t+1}} \right] \end{aligned}$$

expanding the expectation terms and solving yields

$$\begin{aligned}
& \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + 2B_2^\pi \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(b_{42} \varepsilon_{t+1}^z + b_{43} \varepsilon_{t+1}^d + b_{44} \varepsilon_{t+1}^m + b_{45} \varepsilon_{t+1}^s + b_{46} \varepsilon_{t+1}^a + \varepsilon_{t+1}^z) + 2(b_{22} \varepsilon_{t+1}^z + b_{23} \varepsilon_{t+1}^d + b_{24} \varepsilon_{t+1}^m + b_{25} \varepsilon_{t+1}^s + b_{26} \varepsilon_{t+1}^a)} \right] \\
&= \mathbb{E}_t \left[e^{(1+(1-\tau)b_{43}+2b_{23})\varepsilon_{t+1}^d + ((1-\tau)(b_{42}+1)+2b_{22})\varepsilon_{t+1}^z + ((1-\tau)b_{44}+2b_{24})\varepsilon_{t+1}^m + ((1-\tau)b_{45}+2b_{25})\varepsilon_{t+1}^s + ((1-\tau)b_{46}+2b_{26})\varepsilon_{t+1}^a} \right] \\
&= e^{\frac{1}{2} \left((1+(1-\tau)b_{43}+2b_{23})^2 \sigma_d^2 + ((1-\tau)(b_{42}+1)+2b_{22})^2 \sigma_z^2 + ((1-\tau)b_{44}+2b_{24})^2 \sigma_m^2 + ((1-\tau)b_{45}+2b_{25})^2 \sigma_s^2 + ((1-\tau)b_{46}+2b_{26})^2 \sigma_a^2 \right)} \\
\\
& \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) + B_2^\pi \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d + (1-\tau)(b_{42} \varepsilon_{t+1}^z + b_{43} \varepsilon_{t+1}^d + b_{44} \varepsilon_{t+1}^m + b_{45} \varepsilon_{t+1}^s + b_{46} \varepsilon_{t+1}^a + \varepsilon_{t+1}^z) + b_{22} \varepsilon_{t+1}^z + b_{23} \varepsilon_{t+1}^d + b_{24} \varepsilon_{t+1}^m + b_{25} \varepsilon_{t+1}^s + b_{26} \varepsilon_{t+1}^a} \right] \\
&= \mathbb{E}_t \left[e^{(1+(1-\tau)b_{43}+b_{23})\varepsilon_{t+1}^d + ((1-\tau)(b_{42}+1)+b_{22})\varepsilon_{t+1}^z + ((1-\tau)b_{44}+b_{24})\varepsilon_{t+1}^m + ((1-\tau)b_{45}+b_{25})\varepsilon_{t+1}^s + ((1-\tau)b_{46}+b_{26})\varepsilon_{t+1}^a} \right] \\
&= e^{\frac{1}{2} \left((1+(1-\tau)b_{43}+b_{23})^2 \sigma_d^2 + ((1-\tau)(b_{42}+1)+b_{22})^2 \sigma_z^2 + ((1-\tau)b_{44}+b_{24})^2 \sigma_m^2 + ((1-\tau)b_{45}+b_{25})^2 \sigma_s^2 + ((1-\tau)b_{46}+b_{26})^2 \sigma_a^2 \right)}
\end{aligned}$$

Job creation condition. Consider the job creation condition

$$e^{-q_t} = \frac{Q(1-\eta)}{c} ((1 - \gamma e^{\omega_t})e^{z_t} - v) + \beta \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}} \right]$$

the continuation value can be solved in closed form given the linear policies, first note

$$\begin{aligned}
& \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(n_{t+1}+z_{t+1}-n_t-z_t)-q_{t+1}} \right] \\
&= \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}})e^{d_{t+1}-d_t-\tau(B_0^n + B_1^n x_t + B_2^n \varepsilon_{t+1} + z_{t+1} - z_t) - \rho_a a_t - \varepsilon_{t+1}^a + \alpha B_0^\theta + \alpha B_1^\theta x_t + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&= e^{(\rho_d - 1)d_t - \tau(\rho_z - 1)z_t - \tau B_0^n - \tau B_1^n x_t - \rho_a a_t + \alpha B_0^\theta + \alpha B_1^\theta x_t} \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}})e^{\varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right]
\end{aligned}$$

The remaining expectation term can be written as

$$\begin{aligned}
& \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}})e^{\varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&\quad - \delta e^{\rho_s s_t} \mathbb{E}_t \left[e^{\varepsilon_{t+1}^s + \varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right]
\end{aligned}$$

where

$$\begin{aligned}
& \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^d - \tau(b_{42} \varepsilon_{t+1}^z + b_{43} \varepsilon_{t+1}^d + b_{44} \varepsilon_{t+1}^m + b_{45} \varepsilon_{t+1}^s + b_{46} \varepsilon_{t+1}^a + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha(b_{12} \varepsilon_{t+1}^z + b_{13} \varepsilon_{t+1}^d + b_{14} \varepsilon_{t+1}^m + b_{15} \varepsilon_{t+1}^s + b_{16} \varepsilon_{t+1}^a)} \right] \\
&= \mathbb{E}_t \left[e^{(\alpha b_{13} - \tau b_{43} + 1) \varepsilon_{t+1}^d + (\alpha b_{12} - \tau b_{42}) \varepsilon_{t+1}^z + (\alpha b_{14} - \tau b_{44}) \varepsilon_{t+1}^m + (\alpha b_{15} - \tau b_{45}) \varepsilon_{t+1}^s + (\alpha b_{16} - \tau b_{46} - 1) \varepsilon_{t+1}^a} \right] \\
&= e^{\frac{1}{2} \left((\alpha b_{13} - \tau b_{43} + 1)^2 \sigma_d^2 + (\alpha b_{12} - \tau b_{42})^2 \sigma_z^2 + (\alpha b_{14} - \tau b_{44})^2 \sigma_m^2 + (\alpha b_{15} - \tau b_{45})^2 \sigma_s^2 + (\alpha b_{16} - \tau b_{46} - 1)^2 \sigma_a^2 \right)} \\
\\
& \mathbb{E}_t \left[e^{\varepsilon_{t+1}^s + \varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&= \mathbb{E}_t \left[e^{\varepsilon_{t+1}^s + \varepsilon_{t+1}^d - \tau(b_{42} \varepsilon_{t+1}^z + b_{43} \varepsilon_{t+1}^d + b_{44} \varepsilon_{t+1}^m + b_{45} \varepsilon_{t+1}^s + b_{46} \varepsilon_{t+1}^a + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha(b_{12} \varepsilon_{t+1}^z + b_{13} \varepsilon_{t+1}^d + b_{14} \varepsilon_{t+1}^m + b_{15} \varepsilon_{t+1}^s + b_{16} \varepsilon_{t+1}^a)} \right] \\
&= \mathbb{E}_t \left[e^{(\alpha b_{13} - \tau b_{43} + 1) \varepsilon_{t+1}^d + (\alpha b_{12} - \tau b_{42}) \varepsilon_{t+1}^z + (\alpha b_{14} - \tau b_{44}) \varepsilon_{t+1}^m + (\alpha b_{15} - \tau b_{45} + 1) \varepsilon_{t+1}^s + (\alpha b_{16} - \tau b_{46} - 1) \varepsilon_{t+1}^a} \right] \\
&= e^{\frac{1}{2} \left((\alpha b_{13} - \tau b_{43} + 1)^2 \sigma_d^2 + (\alpha b_{12} - \tau b_{42})^2 \sigma_z^2 + (\alpha b_{14} - \tau b_{44})^2 \sigma_m^2 + (\alpha b_{15} - \tau b_{45} + 1)^2 \sigma_s^2 + (\alpha b_{16} - \tau b_{46} - 1)^2 \sigma_a^2 \right)}
\end{aligned}$$

so the expectation term becomes

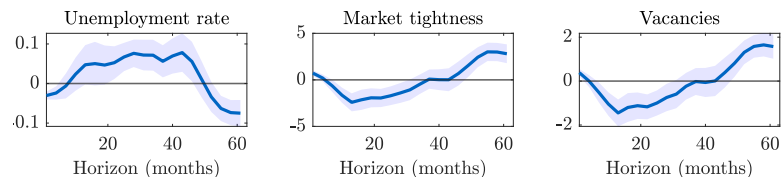
$$\begin{aligned}
& \mathbb{E}_t \left[(1 - \delta e^{s_{t+1}}) e^{\varepsilon_{t+1}^d - \tau(B_2^n \varepsilon_{t+1} + \varepsilon_{t+1}^z) - \varepsilon_{t+1}^a + \alpha B_2^\theta \varepsilon_{t+1}} \right] \\
&= \left(e^{\frac{1}{2} (\alpha b_{15} - \tau b_{45})^2 \sigma_s^2} - \delta e^{\rho_s s_t + \frac{1}{2} (\alpha b_{15} - \tau b_{45} + 1)^2 \sigma_s^2} \right) \\
&\quad \cdot e^{\frac{1}{2} \left((\alpha b_{13} - \tau b_{43} + 1)^2 \sigma_d^2 + (\alpha b_{12} - \tau b_{42})^2 \sigma_z^2 + (\alpha b_{14} - \tau b_{44})^2 \sigma_m^2 + (\alpha b_{15} - \tau b_{45} + 1)^2 \sigma_s^2 + (\alpha b_{16} - \tau b_{46} - 1)^2 \sigma_a^2 \right)}
\end{aligned}$$

Law of motion for employment. Taking the law of motion for employment and substituting the expressions for the policies yields

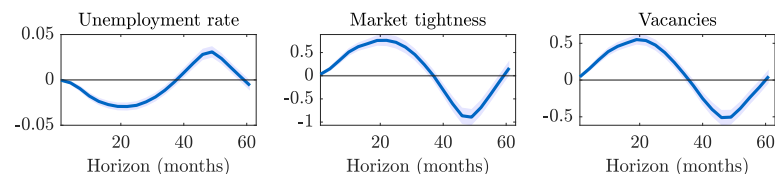
$$\begin{aligned}
& e^{b_{40} + (b_{41} - 1)n_{t-1} + b_{42}z_t + b_{43}d_t + b_{44}m_t + b_{45}s_t + b_{46}a_t} = \\
& 1 - \delta e^{s_t} + Q\theta e^{a_t + (1-\alpha)(b_{10} + b_{11}n_{t-1} + b_{12}z_t + b_{13}d_t + b_{14}m_t + b_{15}s_t + b_{16}a_t)} \left(\frac{e^{-n_{t-1}}}{N} - 1 + \delta e^{s_t} \right)
\end{aligned}$$

E Supporting Empirical Evidence

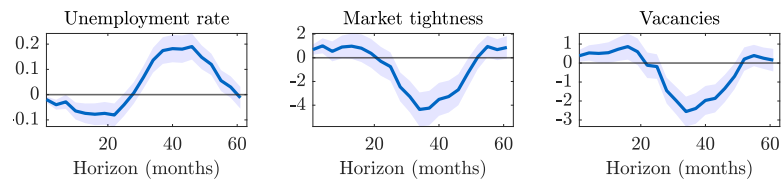
(a) Local Projection Baseline – Romer and Romer (2004) shock, Corporate Profit Proxy, 90% CI



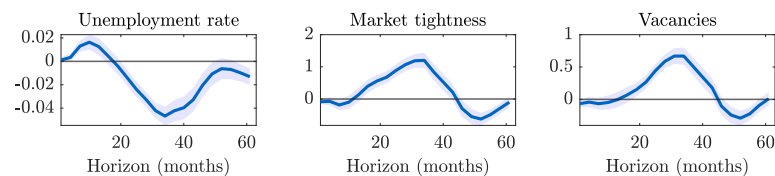
(b) Local Projection Interaction Term – Romer and Romer (2004) shock, Corporate Profit Proxy, 90% CI



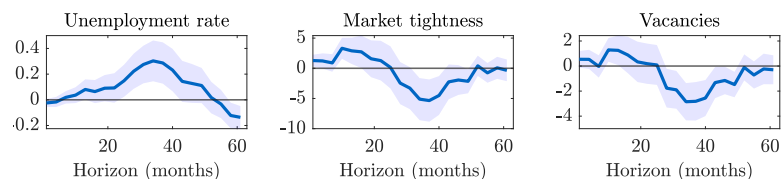
(c) Local Projection Baseline – Aruoba and Drechsel (2024) shock, Corporate Profit Proxy, 90% CI



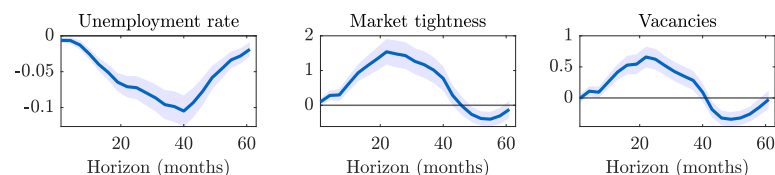
(d) Local Projection Interaction Term – Aruoba and Drechsel (2024) shock, Corporate Profit Proxy, 90% CI



(e) Local Projection Baseline – Bauer and Swanson (2023) shock, Corporate Profit Proxy, 90% CI



(f) Local Projection Interaction Term – Bauer and Swanson (2023) shock, Corporate Profit Proxy, 90% CI



F Stochastic Growth Model with Alternative Stochastic Processes

F.1 Basic stochastic growth model

We illustrate how to accommodate two alternative stochastic processes within our methodology: (i) stochastic volatility and (ii) regime-switching. To simplify exposition, we work through a small example: the stochastic growth model. For any exogenous driving process for total factor productivity, the equilibrium for this model can be summarized as:

$$\begin{aligned} C_t^{-\tau} &= \beta \mathbb{E}_t \left[C_{t+1}^{-\tau} (\alpha e^{z_{t+1}} K_{t+1}^{\alpha-1} + 1 - \delta) \right] \\ K_{t+1} &= e^{z_t} K_t^\alpha + (1 - \delta) K_t - C_t. \end{aligned}$$

In a deterministic steady state, $z = 0$, and the solution is

$$\begin{aligned} K &= \left[\frac{1}{\alpha} \left(\frac{1}{\beta} - 1 + \delta \right) \right]^{\frac{1}{\alpha-1}} \\ C &= K^\alpha - \delta K \end{aligned}$$

In what follows we consider log deviations from steady state value $c_t = \log(C_t/C)$ and $k_t = \log(K_t/K)$, so that $C_t = C e^{c_t}$ and $K_t = K e^{k_t}$. Substitution of this change of variables into the equilibrium gives:

$$\begin{aligned} e^{-\tau c_t} &= \mathbb{E}_t \left[e^{-\tau c_{t+1}} \left\{ (1 - \beta(1 - \delta)) e^{z_{t+1} + (\alpha-1)k_{t+1}} + \beta(1 - \delta) \right\} \right] \\ e^{k_{t+1}} &= \frac{1}{\alpha} \left(\frac{1}{\beta} - (1 - \delta) \right) e^{z_t + \alpha k_t} + (1 - \delta) e^{k_t} - \left(\frac{1}{\alpha} \left(\frac{1}{\beta} - (1 - \delta) \right) - \delta \right) e^{c_t} \end{aligned}$$

Under the assumption that productivity follows an AR(1) in logs

$$z_{t+1} = \rho_z z_t + \sigma_z \epsilon_{t+1}, \quad \epsilon_t \sim N(0, 1),$$

the linear policies are given by

$$\begin{aligned} c_t &= a_0 + a_1 k_t + a_2 z_t \\ k_{t+1} &= b_0 + b_1 k_t + b_2 z_t \end{aligned}$$

which implies

$$\begin{aligned} c_{t+1} &= a_0 + a_1 b_0 + a_1 b_1 k_t + a_1 b_2 z_t + a_2 z_{t+1} \\ &= a_0 + a_1 b_0 + a_1 b_1 k_t + a_1 b_2 z_t + a_2 (\rho_z z_t + \sigma_z \epsilon_{t+1}) \\ &= a_0 + a_1 b_0 + a_1 b_1 k_t + (a_1 b_2 + a_2 \rho_z) z_t + a_2 \sigma_z \epsilon_{t+1} \end{aligned}$$

Returning to the expectation operator, under these restrictions, we have:

$$\begin{aligned} & \mathbb{E}_t \left[e^{-\tau c_{t+1}} \left\{ (1 - \beta(1 - \delta)) e^{z_{t+1} + (\alpha-1)k_{t+1}} + \beta(1 - \delta) \right\} \right] \\ &= (1 - \beta(1 - \delta)) \mathbb{E}_t \left[e^{-\tau c_{t+1} + z_{t+1} + (\alpha-1)k_{t+1}} \right] + \beta(1 - \delta) \mathbb{E}_t \left[e^{-\tau c_{t+1}} \right] \end{aligned}$$

The first term becomes

$$\begin{aligned} & \mathbb{E}_t \left[e^{-\tau c_{t+1} + z_{t+1} + (\alpha-1)k_{t+1}} \right] \\ &= e^{-\tau(a_0 + a_1 b_0 + a_1 b_1 k_t + (a_1 b_2 + a_2 \rho_z) z_t) + \rho_z z_t + (\alpha-1)k_{t+1}} \mathbb{E}_t \left[e^{(1-\tau a_2) \sigma_z \epsilon_{t+1}} \right] \\ &= e^{-\tau(a_0 + a_1 b_0 + a_1 b_1 k_t + (a_1 b_2 + a_2 \rho_z) z_t) + \rho_z z_t + (\alpha-1)k_{t+1} + \frac{(1-\tau a_2)^2 \sigma_z^2}{2}} \end{aligned}$$

and the second term

$$\mathbb{E}_t \left[e^{-\tau c_{t+1}} \right] = e^{-\tau(a_0 + a_1 b_0 + a_1 b_1 k_t + (a_1 b_2 + a_2 \rho_z) z_t) + \frac{(\tau a_2)^2 \sigma_z^2}{2}}.$$

F.2 Stochastic Volatility

Stochastic volatility models have an exogenous shock process that no longer maps into a VAR-process as in Equation (2). We can still use (dynamic) Taylor projections to solve the model, as we will elaborate on below, but we need to make adaptations to the filter.

The attractive feature of dynamic Taylor projection with stochastic volatility is that the time-varying risk introduced by the stochastic volatility term affects policy functions, because the model solution is not certainty equivalent. Following De Groot (2016), if one would use perturbation methods instead, one needs at least a fourth or sixth order perturbation for the variance of the volatility shocks to show up in the coefficients of the decision rules.

For exposition, assume our model is the standard stochastic growth model, augmented with a shock process with stochastic volatility (SV), that is, TFP is z_t and follows an AR(1)-SV:

$$\begin{aligned} z_t &= \rho_z z_{t-1} + e^{\xi_t} u_t \\ \xi_t &= (1 - \rho_\xi) \mu_\xi + \rho_\xi \xi_{t-1} + v_t \end{aligned}$$

In that case, we have three states: capital, TFP (z_t) and volatility ξ_t , such that our policy rule for consumption resulting from Taylor Projection takes the form:

$$c_t = a_{0,x} + a_{1,x} k_t + a_{2,x} z_t + a_{3,x} \xi_t$$

and our linearized law of motion for capital is given by:

$$k_{t+1} = b_{0,x} + b_{1,x}k_t + b_{2,x}z_t + b_{3,x}\xi_t$$

Expectations Returning to the previous expectation terms in the Euler equation, we have the first term:

$$\begin{aligned} & \mathbb{E}_t \left[e^{-\tau c_{t+1} + z_{t+1} + (\alpha-1)k_{t+1}} \right] \\ &= \mathbb{E}_t \left[e^{-\tau(a_0 + a_1 k_{t+1} + a_2 z_{t+1} + a_3 x_{t+1}) + z_{t+1} + (\alpha-1)k_{t+1}} \right] \\ &= e^{-\tau(a_0 + a_1 k_{t+1}) + (\alpha-1)k_{t+1}} \mathbb{E}_t \left[e^{-\tau(a_2 z_{t+1} + a_3 x_{t+1}) + z_{t+1}} \right] \\ &= e^{-\tau(a_0 + a_1 k_{t+1}) + (\alpha-1)k_{t+1}} \mathbb{E}_t \left[e^{-\tau \left(a_2 \left(\rho_z z_t + e^{\frac{x_{t+1}}{2}} \epsilon_{t+1} \right) + a_3 \left((1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1} \right) \right) + \rho_z z_t + e^{\frac{x_{t+1}}{2}} \epsilon_{t+1}} \right] \end{aligned}$$

Note, as k_{t+1} is determined by time t , we can ignore it for the purposes of characterizing the expectation, so we have

$$\begin{aligned} & \mathbb{E}_t \left[e^{-\tau \left(a_2 \left(\rho_z z_t + e^{\frac{x_{t+1}}{2}} \epsilon_{t+1} \right) + a_3 \left((1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1} \right) \right) + \rho_z z_t + e^{\frac{x_{t+1}}{2}} \epsilon_{t+1}} \right] \\ &= e^{(1-\tau a_2)\rho_z z_t - \tau a_3((1-\rho_x)\mu + \sigma_x x_t)} \mathbb{E}_t \left[e^{\left\{ (1-\tau a_2) e^{\frac{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}}{2}} \epsilon_{t+1} - \tau a_3 \sigma_x \eta_{t+1} \right\}} \right] \end{aligned}$$

Again, there are terms that are determined at time t , which do not influence the expectation. The remaining expression is an expectation over η_{t+1} and ϵ_{t+1} , but application of iterated expectation will show that $\epsilon_{t+1}|\eta_{t+1}$ can be solved analytically, using the properties of Gaussian distributions, as:

$$\begin{aligned} & \mathbb{E}_t \left[e^{\left\{ (1-\tau a_2) e^{\frac{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}}{2}} \epsilon_{t+1} - \tau a_3 \sigma_x \eta_{t+1} \right\}} \right] \\ &= \mathbb{E}_{\eta_{t+1}} \left[\mathbb{E}_{\epsilon_{t+1}} \left[e^{\left\{ (1-\tau a_2) e^{\frac{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}}{2}} \epsilon_{t+1} - \tau a_3 \sigma_x \eta_{t+1} \right\}} \right] \middle| \eta_{t+1} \right] \\ &= \mathbb{E}_{\eta_{t+1}} \left[e^{-\tau a_3 \sigma_x \eta_{t+1}} \mathbb{E}_{\epsilon_{t+1}} \left[e^{\left\{ (1-\tau a_2) e^{\frac{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}}{2}} \epsilon_{t+1} \right\}} \right] \middle| \eta_{t+1} \right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}_{\eta_{t+1}} \left[e^{-\tau a_3 \sigma_x \eta_{t+1}} e^{\left\{ \frac{(1-\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}} \right\}} \right] \\
&= \mathbb{E}_{\eta_{t+1}} \left[e^{-\tau a_3 \sigma_x \eta_{t+1} + \left\{ \frac{(1-\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}} \right\}} \right] \\
&= \int_{-\infty}^{\infty} \left\{ e^{-\tau a_3 \sigma_x \eta_{t+1} + \left\{ \frac{(1-\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}} \right\}} \right\} \frac{1}{\sqrt{2\pi}} e^{-\frac{\eta_{t+1}^2}{2}} d\eta_{t+1}
\end{aligned}$$

While this expression does not have a closed-form in general, it can be well approximated by a Gauss-Hermite quadrature over η_{t+1} , so the expression becomes

$$\sum_{i=1}^N \left\{ e^{-\tau a_3 \sigma_x \eta_i + \left\{ \frac{(1-\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_i} \right\}} \right\} \omega_i$$

where $\{\eta_i, \omega_i\}_{i=1}^N$ are the nodes and weights. The second term in the expectation becomes:

$$\begin{aligned}
\mathbb{E}_t [e^{-\tau c_{t+1}}] &= \mathbb{E}_t [e^{-\tau(a_0 + a_1 k_{t+1} + a_2 z_{t+1} + a_3 x_{t+1})}] \\
&= e^{-\tau(a_0 + a_1 k_{t+1})} \mathbb{E}_t [e^{-\tau(a_2 z_{t+1} + a_3 x_{t+1})}] \\
&= e^{-\tau(a_0 + a_1 k_{t+1} + a_2 \rho_z z_t + a_3 [(1-\rho_z)\mu + \rho_x x_t])} \mathbb{E}_t \left[e^{-\tau a_2 e^{\frac{x_{t+1}}{2}} \epsilon_{t+1} - \tau a_3 \sigma_x \eta_{t+1}} \right]
\end{aligned}$$

Using iterated expectations as before, we have:

$$\mathbb{E}_t \left[e^{-\tau a_2 e^{\frac{x_{t+1}}{2}} \epsilon_{t+1} - \tau a_3 \sigma_x \eta_{t+1}} \right] = E_{\eta_{t+1}} \left[e^{\frac{(\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_{t+1}} - \tau a_3 \sigma_x \eta_{t+1}} \right]$$

which can be approximated as

$$\sum_{i=1}^n \left\{ e^{\frac{(\tau a_2)^2}{2} e^{(1-\rho_x)\mu + \rho_x x_t + \sigma_x \eta_i} - \tau a_3 \sigma_x \eta_i} \right\} \omega_i.$$

State Space Form Assume we observe both output, given by $y_t = z_t + \alpha k_t$ and consumption. In that case, we obtain a measurement equation

$$\begin{bmatrix} y_t \\ c_t \end{bmatrix} = \begin{bmatrix} 0 \\ \alpha_{0,x} \end{bmatrix} + \begin{bmatrix} \alpha & 1 & 0 \\ a_{1,x} & a_{2,x} & a_{3,x} \end{bmatrix} \begin{bmatrix} k_t \\ z_t \\ \xi_t \end{bmatrix} + \varepsilon_t, \quad \varepsilon_t \sim N(0, H)$$

with H measurement error variance. Labeling $x_t = (k_t, z_t, \xi_t)'$, and $\eta_t = (u_t, v_t)'$, we obtain a state transition equation of the form:

$$x_t = \begin{bmatrix} b_{0,x} \\ 0 \\ (1 - \rho_\xi)\mu_\xi \end{bmatrix} + \begin{bmatrix} b_{1,x} & b_{2,x} & b_{3,x} \\ 0 & \rho_z & 0 \\ 0 & 0 & \rho_\xi \end{bmatrix} x_{t-1} + \underbrace{\begin{bmatrix} 0 & 0 \\ e^{\xi_t} & 0 \\ 0 & 1 \end{bmatrix}}_{R(x_t)} \eta_t, \quad \eta_t \sim N(0, \text{diag}(\sigma_u, \sigma_v)^2).$$

Filtering The only deviation from our standard framework presented above is that $R(x_t)$ is now a nonlinear function. The most computationally efficient way to handle this nonlinearity is to use the Extended Kalman filter, which replaces $R(x_t)$ by $R(\hat{x}_{t|t-1})$, i.e., a zeroth-order Taylor expansion around the last state forecast. In our case, that means replacing $\begin{bmatrix} 0 & e^{\xi_t} & 1 \end{bmatrix}$ by $\begin{bmatrix} 0 & e^{\hat{\xi}_{t|t-1}} & 1 \end{bmatrix}$.

If one does not like EKF, since this is an approximate filter, a second approach that is easy to implement follows from the idea in Friedman and Harris (1998), to discretize the stochastic process for the volatility term ξ_t using finite-state Markov chain approximations, which one can do with, for example, the Rouwenhorst (1995) or Tauchen (1986) method. Conditional on ξ_t , the state space model is Gaussian and linear. This allows one to then use the Hamilton (1994) filter to evaluate the likelihood. This builds on seminal work by Kitagawa (1996). We do not separately discuss this approach, as it is nested in the next application, which considers regime-switching.

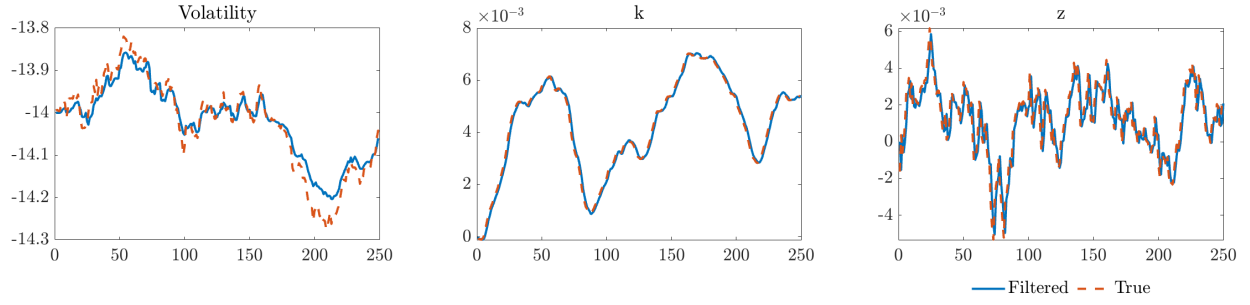
Demonstration Figures F1 and F2 show the performance of both filters. We simulate a dataset of length $T = 250$ of both output and consumption. For the discretization-based filter, we use the Tauchen method with 250 grid points. In order to be able to well identify the volatility component, it is important to have little measurement error. Hence, we assume there is no measurement error on consumption in this simulation, and 0.045% measurement error on output. As the figures show, both filters track capital and TFP well. Volatility has small loadings in the policy rules, which implies it is hard to identify, but also that the ability to filter it well has little effect on the fitted values in Figure F2, which both look good. The discretization filter is more volatile than the EKF but captures the trough around $T = 200$ better and does a better job tracking k and z .

F.3 Regime Switching

Another variation of a stochastic process regularly encountered in the literature is to assume the dynamics of the latent states are governed by a regime-switching process. This nests models that have recently gained popularity such as rare disaster models. More specifically, assume there is a latent variable $s_t \in \{0, 1\}$ whose dynamics are given by $p(s_{t+1} = 1 | s_t = 1) = p_1$ and

Figure F1: Filtered States

(a) Extended Kalman filter



(b) Discretized volatility following Friedman and Harris (1998)

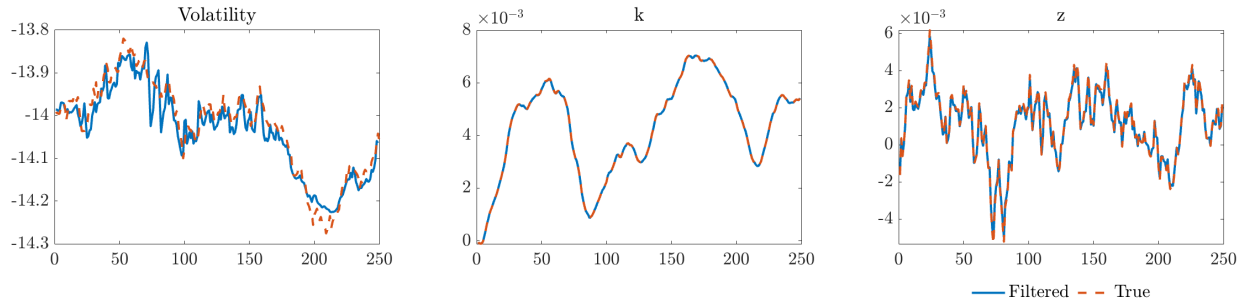
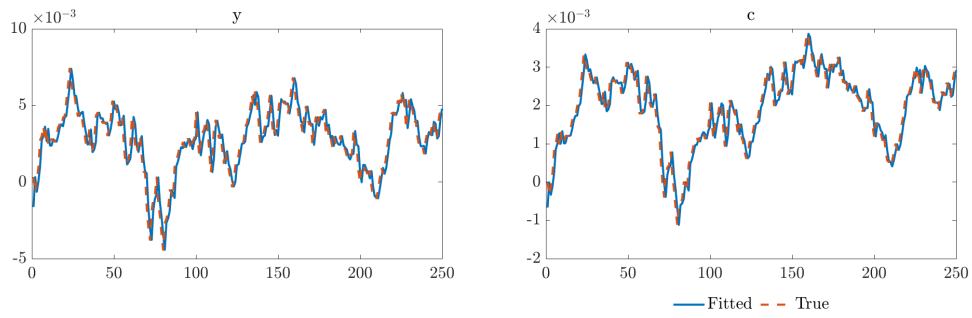
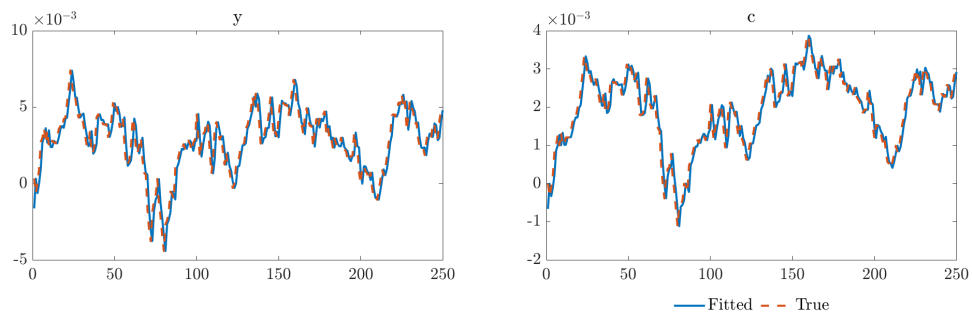


Figure F2: Fitted Values

(a) Extended Kalman filter



(b) Discretized volatility following Friedman and Harris (1998)



$p(s_{t+1} = 0 | s_t = 0) = p_0$, that is,

$$s_t \sim \Pi, \quad \Pi = \begin{bmatrix} p_0 & 1 - p_0 \\ 1 - p_1 & p_1 \end{bmatrix}.$$

Note that the framework can be extended for an arbitrary number of finite states, but for ease of exposition, in this example, we use two states. Latent state s_t governs the dynamics of z_t , in the following way:

$$z_{t+1} = \begin{cases} \rho_0 z_t + \sigma_0 \eta_t & \text{if } s_t = 0 \\ \rho_1 z_t + \sigma_1 \eta_t & \text{if } s_t = 1 \end{cases} \quad \eta_t \sim N(0, 1) \quad (\text{F.1})$$

As above, $y_t = z_t + \alpha k_t$. Collect $x_t = (k_t, z_t)$. Our policy rule for consumption now takes the form:

$$c_t = \begin{cases} a_{0,0,x} + a_{1,0,x} k_t + a_{2,0,x} z_t & \text{if } s_t = 0 \\ a_{0,1,x} + a_{1,1,x} k_t + a_{2,1,x} z_t & \text{if } s_t = 1 \end{cases}$$

and our linearized law of motion for capital:

$$k_{t+1} = \begin{cases} b_{0,0,x} + b_{1,0,x} k_t + b_{2,0,x} z_t & \text{if } s_t = 0 \\ b_{0,1,x} + b_{1,1,x} k_t + b_{2,1,x} z_t & \text{if } s_t = 1 \end{cases}$$

Expectations These policies imply that the choice of consumption c_{t+1} depends on the path of states s_t, s_{t+1} as:

$$c_{t+1} = \begin{cases} a_{00} + a_{01} b_{00} + a_{01} b_{01} k_t + (a_{01} b_{02} + a_{02} \rho_0) z_t + a_{02} \sigma_0 \epsilon_{t+1}, & \text{if } s_t = 0, s_{t+1} = 0 \\ a_{10} + a_{11} b_{00} + a_{11} b_{01} k_t + (a_{11} b_{02} + a_{12} \rho_1) z_t + a_{12} \sigma_1 \epsilon_{t+1}, & \text{if } s_t = 0, s_{t+1} = 1 \\ a_{00} + a_{01} b_{10} + a_{01} b_{11} k_t + (a_{01} b_{12} + a_{02} \rho_0) z_t + a_{02} \sigma_0 \epsilon_{t+1}, & \text{if } s_t = 1, s_{t+1} = 0 \\ a_{10} + a_{11} b_{10} + a_{11} b_{11} k_t + (a_{11} b_{12} + a_{12} \rho_1) z_t + a_{12} \sigma_1 \epsilon_{t+1}, & \text{if } s_t = 1, s_{t+1} = 1 \end{cases}$$

The expectation terms in the Markov model can be written as

$$\mathbb{E}_t[\cdot] = \mathbb{E}_{s_{t+1}}[\mathbb{E}_{z_{t+1}}[\cdot | s_{t+1}, s_t, z_t] | s_t].$$

Taking the same expectation terms as before, we have

$$\mathbb{E}_{z_{t+1}} \left[e^{-\tau c_{t+1} + z_{t+1} + (\alpha-1)k_{t+1}} \right] = \begin{cases} e^{-\tau(a_{00}+a_{01}b_{00}+a_{01}b_{01}k_t+(a_{01}b_{02}+a_{02}\rho_0)z_t)+\rho_0 z_t+(\alpha-1)k_{t+1}+\frac{(1-\tau a_{02})^2 \sigma_0^2}{2}}, & \text{if } s_t = 0, s_{t+1} = 0 \\ e^{-\tau(a_{10}+a_{11}b_{00}+a_{11}b_{01}k_t+(a_{11}b_{02}+a_{12}\rho_1)z_t)+\rho_1 z_t+(\alpha-1)k_{t+1}+\frac{(1-\tau a_{12})^2 \sigma_1^2}{2}}, & \text{if } s_t = 0, s_{t+1} = 1 \\ e^{-\tau(a_{00}+a_{01}b_{10}+a_{01}b_{11}k_t+(a_{01}b_{12}+a_{02}\rho_0)z_t)+\rho_0 z_t+(\alpha-1)k_{t+1}+\frac{(1-\tau a_{02})^2 \sigma_0^2}{2}}, & \text{if } s_t = 1, s_{t+1} = 0 \\ e^{-\tau(a_{10}+a_{11}b_{10}+a_{11}b_{11}k_t+(a_{11}b_{12}+a_{12}\rho_1)z_t)+\rho_1 z_t+(\alpha-1)k_{t+1}+\frac{(1-\tau a_{12})^2 \sigma_1^2}{2}}, & \text{if } s_t = 1, s_{t+1} = 1 \end{cases}$$

where it should be noted that k_{t+1} depends on the initial states s_t , and

$$\mathbb{E}_{z_{t+1}} [e^{-\tau c_{t+1}}] = \begin{cases} e^{-\tau(a_{00}+a_{01}b_{00}+a_{01}b_{01}k_t+(a_{01}b_{02}+a_{02}\rho_0)z_t)+\frac{(\tau a_{02})^2 \sigma_0^2}{2}}, & \text{if } s_t = 0, s_{t+1} = 0 \\ e^{-\tau(a_{10}+a_{11}b_{00}+a_{11}b_{01}k_t+(a_{11}b_{02}+a_{12}\rho_1)z_t)+\frac{(\tau a_{12})^2 \sigma_1^2}{2}}, & \text{if } s_t = 0, s_{t+1} = 1 \\ e^{-\tau(a_{00}+a_{01}b_{10}+a_{01}b_{11}k_t+(a_{01}b_{12}+a_{02}\rho_0)z_t)+\frac{(\tau a_{02})^2 \sigma_0^2}{2}}, & \text{if } s_t = 1, s_{t+1} = 0 \\ e^{-\tau(a_{10}+a_{11}b_{10}+a_{11}b_{11}k_t+(a_{11}b_{12}+a_{12}\rho_1)z_t)+\frac{(\tau a_{12})^2 \sigma_1^2}{2}}, & \text{if } s_t = 1, s_{t+1} = 1 \end{cases}$$

This implies that also for this model, we do not need numerical integration to compute the Taylor Projection solution.

State Space Representation Conditional on s_t , this is simply a linear state space system, with, for $s_t = j$, a measurement equation

$$\begin{bmatrix} y_t \\ c_t \end{bmatrix} = \begin{bmatrix} 0 \\ a_{0,j,x} \end{bmatrix} + \begin{bmatrix} \alpha & 1 \\ a_{1,j,x} & a_{2,j,x} \end{bmatrix} \begin{bmatrix} k_t \\ z_t \end{bmatrix} + \varepsilon_t, \quad \varepsilon_t \sim N(0, H).$$

and, again for $s_t = j$, the state transition equation is given by

$$\begin{bmatrix} k_{t+1} \\ z_{t+1} \end{bmatrix} = \begin{bmatrix} b_{0,j,x} \\ 0 \end{bmatrix} + \begin{bmatrix} b_{1,j,x} & b_{2,j,x} \\ 0 & \rho_j \end{bmatrix} \begin{bmatrix} k_t \\ z_t \end{bmatrix} + \begin{bmatrix} 0 \\ \sigma_j \end{bmatrix} \eta_t, \quad \eta_t \sim N(0, 1).$$

Filter To filter, initialize $x_{1|0}$ and $P_{1|0}$, as well as $\hat{\xi}_{1|0}$ which contain the probabilities $P(s_t = j)$ before observing any data, so, for example, use the stationary distribution implied by Π . Solve the model using Taylor Projections at $x_{1|0}$. Next, do the update and prediction step of Algorithm 1 for $s_t = j$, $j = 0, 1$ separately. Then, compute the likelihood contribution for $j = 0, 1$, based on evaluating the forecast error v_t using the normal distribution with mean zero and variance F_t , which we denote $\ell_t^{(j)}$.

Update the regime probabilities:

$$\hat{\xi}_{t|t}^{(j)} := \mathbb{P}(s_t = j \mid y_{1:t}^{\text{obs}}) = \frac{\ell_t^{(j)} \cdot \hat{\xi}_{t|t-1}^{(j)}}{\sum_{j'=0}^1 \ell_t^{(j')} \cdot \hat{\xi}_{t|t-1}^{(j')}}, \quad j \in \{0, 1\},$$

Next, we follow Hamilton (1994) and compute the predicted regime probabilities:

$$\hat{\xi}_{t|t-1} = \Pi \cdot \xi_{t-1|t-1}, \quad j \in \{0, 1\},$$

where $\hat{\xi}_{t-1|t-1}^{(i)} = \mathbb{P}(s_{t-1} = i \mid y_{1:t-1}^{\text{obs}})$ is the filtered probability from last iteration, and $\hat{\xi}_{t|t-1}^{(i)}$ is the forecast for the probability that $s_t = i$ given all data $y_{1:t-1}^{\text{obs}}$.

Finally, compute the filtered and forecasted estimates of latent variables by taking the regime-weighted average, for example,

$$\hat{z}_{t|t} = \sum_{j=0}^1 \hat{\xi}_{t|t}^{(j)} \cdot \hat{z}_{t|t}^{(j)}, \quad \text{and} \quad \hat{z}_{t+1|t} = \sum_{j=0}^1 \hat{\xi}_{t+1|t}^{(j)} \cdot \hat{z}_{t+1|t}^{(j)},$$

where $\hat{z}_{t|t}^{(j)}$ denotes the filtered state z_t conditional on regime j . As before, we then solve the model using the Taylor Projection method at $\hat{x}_{t+1|t}$, and proceed again with the update and prediction step of the Kalman filter.

Calibration We use the following parameterization: $p_0 = 0.97$, $\rho_0 = 0.8$, $\sigma_0 = 0.01$, $p_1 = 0.95$, $\rho_1 = 0.95$, $\sigma_1 = 0.002$, risk aversion $\tau = 2$, $\beta = 1.04^{-1/4}$, $\alpha = 0.36$ and $\delta = 0.025$.

Demonstration To illustrate the performance of the filter described above, we simulate a dataset of length $T = 250$ from the dynamic Taylor Projection model solution. We assume measurement error is 10% of the standard deviation of the observed data in the simulation below. We then recover the latent states using the filter. Results are visualized in Figure F3 and F4. As we can see in these figures, the filtered states track the true values used in the simulation closely.

Figure F3: Fitted values

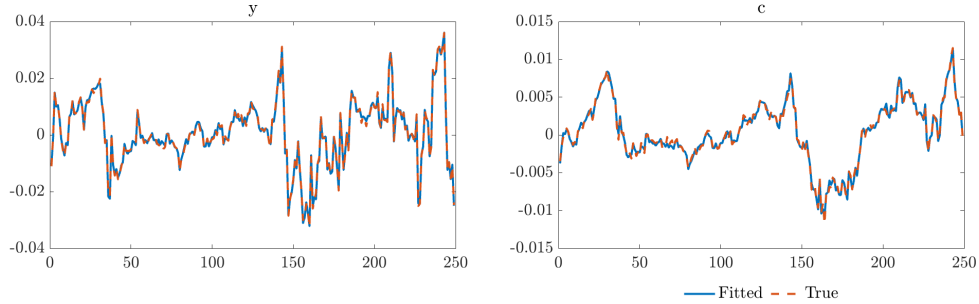
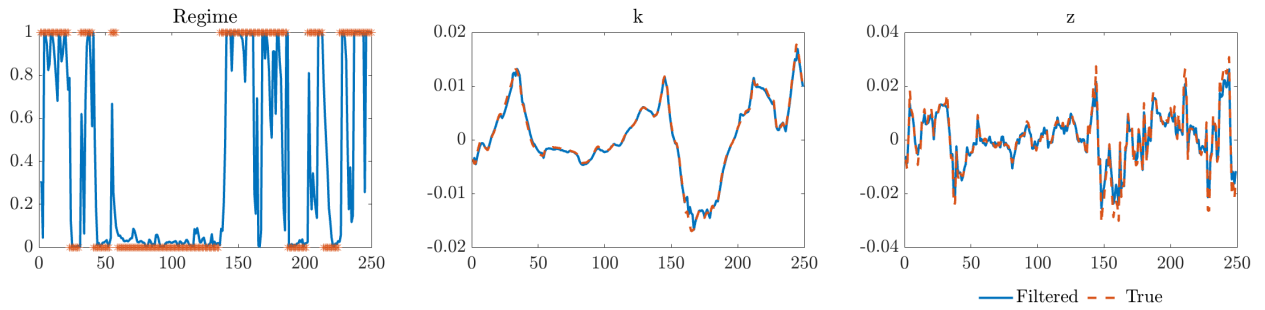


Figure F4: Filtered and true states values



Run time It takes about 1.3 seconds to run the filter 100 times on a dataset of length $T = 250$.

G Heterogeneous Agent Models

We now provide a short description of the Krusell and Smith (1998) model.

Households A continuum of households indexed by $j \in [0, 1]$ have preferences over consumption c_{jt} , represented by

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \frac{c_{jt}^{1-\sigma} - 1}{1-\sigma},$$

where β is the discount factor and $1/\sigma$ is the elasticity of intertemporal substitution. Labor is supplied inelastically, and each household's employment status $\epsilon_{jt} \in \{0, 1\}$ evolves according to transition probabilities $\pi(\epsilon' | \epsilon; z')$, which depend on aggregate TFP z . Employed households earn after-tax wages $w_t(1 - \tau_t)$, while the unemployed receive benefits bw_t . Assuming a balanced government budget in every period implies that the labor income tax rate satisfies $\tau_t = b(1 - L_t)/L_t$, where L_t is the aggregate employment rate. Households hold assets a_{jt} paying real return r_t , but face a borrowing constraint $a_{jt+1} \geq 0$.

Firm We assume a representative firm, producing output according to production function $Y_t = e^{z_t} K_t^\alpha L_t^{1-\alpha}$ with z_t aggregate TFP, which follows an AR(1), K_t aggregate capital, and α the capital share. Capital depreciates at rate δ and costs r_t , a unit of labor costs w_t .

Equilibrium The aggregate states of the model are $x_t = (z_t, \mu_t)$, with μ_t the distribution of households over employment and assets, i.e., (ϵ, a) . The policies are next period's assets, $a'(\epsilon, a; z, \mu)$, interest rate $r(z, \mu)$, wages $w(z, \mu)$ and distribution $\mu'(z, \mu)$, which have to satisfy three sets of conditions:

1. For given prices and law of motion of the distribution, household optimization satisfies: $c(\epsilon, a; z, \mu)^{-\sigma} \geq \beta \mathbb{E} [(1 + r(z', \mu u'(z, \mu))) c(\epsilon', a'(\epsilon, a; z, \mu); z', \mu'(z, \mu))^{-\sigma} | \epsilon, z, \mu]$ with equality if $a'(\epsilon, a; z, \mu) \geq 0$. Here $c(\epsilon, a; z, \mu) = w(z, \mu)((1 - \tau)\epsilon + b(1 - \epsilon)) + (1 + r(z, \mu))a - a'(\epsilon, a; z, \mu)$ is optimal consumption.
2. Interest rate and wages satisfy:

$$\begin{aligned} r(z, \mu) &= \alpha e^z K^{\alpha-1} L^{1-\alpha} - \delta \\ w(z, \mu) &= (1 - \alpha) e^z K^\alpha L^{-\alpha} \end{aligned}$$

where $K = \sum_{\epsilon} \int a d\mu(\epsilon, a)$.

3. Law of motion for the distribution: for all measurable sets Λ_a ,

$$\mu'(z, \mu)(\epsilon, \Lambda_a) = \sum_{\tilde{\epsilon}} \pi(\epsilon | \tilde{\epsilon}, z) \int 1\{a'(\tilde{\epsilon}, a; z, \mu) \in \Lambda_a\} \mu(\tilde{\epsilon}, da)$$